

Table of Contents

1. Executive summary	2
2. Introduction	3
3. Understanding DePIN & Its Relevance	4
3.1 Definition & Taxonomy	4
3.2 Economic Rationale	5
3.3. Progress on DePIN: A Timeline of Key Milestones	6
3.3 Market Traction	7
4. Why Solana? Technical and Strategic Fit for DePIN	8
5. Demand and Supply Resonance Framework	11
5.1 Supply-side growth - how Solana is onboarding real infrastructure	12
5.2. Demand-side & monetisation - where the cash actually lands	14
5.3 Supply-Demand Synthesis	16
6. Current State of DePIN on Solana - Sector deep dives	18
6.1. Wireless Networks (DeWi)	18
6.2. Storage Networks	22
6.3. Sensor Networks	23
6.4. Compute Networks and AI	26
6.5. Services	29
*6.6. DePIN Infrastructure Tools	32
Cross-sector conclusions	33
7. Capital flow	34
Overview	34
Venture momentum & stage-mix	35
Key take-aways	36
8. Winning Entities and Growth Analysis	37
8.1. Verticals and Winning Strategies	38
9. Recent News and Developments	39
10. Challenges and Risks	39
11. Outlook & & Emerging Primitives	40
Expansion into Emerging DePIN Sectors	40
Scenario glide-path (base vs bull)	45
Bottom line	46
12. Conclusion and Final Remarks: The State of the Solana DePIN Ecosystem	46
13. References and Data Sources	47

1. Executive summary

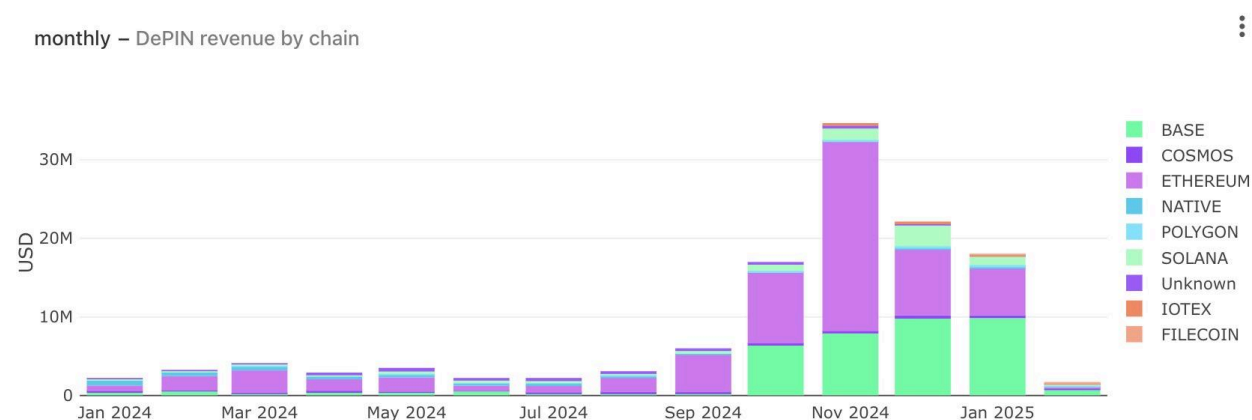
- The Solana Decentralized Physical Infrastructure Networks (DePIN) ecosystem has emerged as a transformative force in the blockchain landscape, leveraging its high-performance infrastructure to tokenize and coordinate real-world assets such as wireless networks, storage systems, sensors, compute resources, and emerging services. As of July 2025, this report - grounded in on-chain data from Dune Analytics, Topledder dashboards, Messari's 2024 DePIN study, and project disclosures - provides a comprehensive, data-driven assessment of Solana's DePIN sector, capturing its current state, growth dynamics, and future potential.
- **Market Leadership and Scale:** Solana commands a 39.3% share of the global DePIN market, with a total ecosystem valuation of \$3.25 billion and \$5.9 million in cumulative onchain revenue (Dune, April 2025). This dominance, despite hosting only 5% of listed DePIN projects, is driven by its technical edge - 65,000 transactions per second (TPS), 400 ms block finality, and sub-cent fees (\$0.00025) - making it ideal for high-frequency, micro-payment-intensive workloads. Key projects like Helium (\$2.3M revenue, 176K+ subscribers), Hivemapper (77K+ nodes, \$50K+ peak revenue), and Render Network (\$300K/week peak) lead a robust network of 238,000+ registered nodes, with a 20% quarterly adoption increase evidenced by milestones such as Lyft's Bee Maps integration (May 2025) and Nosana's node doubling to 600 daily post-mainnet launch (January 2025).
- **Sector Insights:** The ecosystem spans five core verticals - Wireless Networks (\$7B market, led by Helium), Storage Networks (\$200M, driven by shdwDrive), Sensor Networks (\$1.5B, anchored by Hivemapper), Compute Networks and AI (\$500M, dominated by Render), and the nascent Services sector (\$100M) - each addressing trillion-dollar total addressable markets (TAM) with compound annual growth rates (CAGR) exceeding 15% (Statista, Messari, 2025). Solana's infrastructure tools, including Sphere Gateway (\$6.3M in top-ups), cNFTs (1.02M Helium hotspots), and the Solana Mobile Stack (55,000 Saga-2 units shipped), enhance scalability and user adoption, while enterprise partnerships with AT&T, Lyft, and TomTom validate real-world utility.
- **Growth Dynamics:** Supply-side growth is fueled by the Solana Mobile Stack's 54% mobile node share and hardware innovations like Helium's \$249 Wi-Fi hotspots (halving payback periods to 19 months), while demand-side monetization - \$130K weekly revenue peaks - relies on diverse models like Helium's \$20 ARPU subscriptions and Render's \$0.03/GFLOP-hr GPU pricing, undercutting Web2 incumbents by 60%. Capital flows reached \$173M in Q3-2024, with community-funded hardware (\$0.6B cumulative) and high-FDV token launches (e.g., MOBILE at \$1.1B) reinforcing financial moats. The ecosystem's \$12M annual revenue and 3.1x node growth (2024-2025) align with Messari's 100-1,000x growth projection.
- **Future Primitives and Outlook:** Looking to 2026-2027, novel primitives like Decentralized Microgrid Optimization Networks, Adaptive Sensor Fusion Ecosystems, and Tokenized Skill-Based Networks promise to expand TAM to \$30B, with \$10M projected revenue in 2026 (Topledder). These innovations, supported by Solana's sub-cent fees and Token-22 programmability, could drive node counts to 2.5M and

DePIN's share of SOL FDV to 30% in a bull scenario, though regulatory costs (\$500M annually) and EVM competition (46.6% market share) pose risks.

- **Strategic Implications:** For developers, success hinges on leveraging modular tools, piloting cross-sector integrations, and prioritizing demand-side abstraction. Investors should target high-growth verticals while mitigating regulatory and competitive challenges through diversified portfolios. The Solana DePIN ecosystem, with its blend of technical superiority, community engagement, and real-world adoption, is poised to lead Web3's infrastructure revolution, targeting one-third of global DePIN flows by 2027, provided it navigates its evolving risk landscape effectively.

2. Introduction

Decentralized Physical Infrastructure Networks (DePIN) have emerged as a distinct crypto vertical that tokenizes *real-world* infrastructure - 5G radios, dash-cams, GPU clusters - and coordinates their deployment through open, incentive-driven marketplaces. The sector's investable universe ballooned from **≈150 tokens and \$17 bn FDV in 2022 to >350 tokens and \$50 bn FDV by late-2024**, a 2.9 x expansion that outpaced every other crypto sub-sector except AI.



This report - grounded in on-chain telemetry from **Dune**, **TopLedger/Flipside** dashboards, Messari's *State of DePIN 2024* study, and first-party project disclosures - delivers an institutional-grade assessment of the **Solana-centric DePIN landscape as of 5 July 2025**. Specifically, it aims to:

1. **Map the competitive field:** quantify Solana's share of DePIN market cap, node count, and revenue relative to rival L1s.

2. **Identify out-performers:** dissect business models and KPIs of Helium, Hivemapper, Render, Nosana and emerging challengers.
3. **Unpack growth drivers:** analyse token-economic designs, enterprise partnerships, and developer tooling that reinforce network effects.
4. **Stress-test the thesis:** outline technological, regulatory, and competitive risks that could erode Solana's current edge.

The end-goal is a neutral but data-rich framework that LPs, founders, and strategists can use when deciding **where** and **how** to build the next generation of physical-infrastructure protocols.

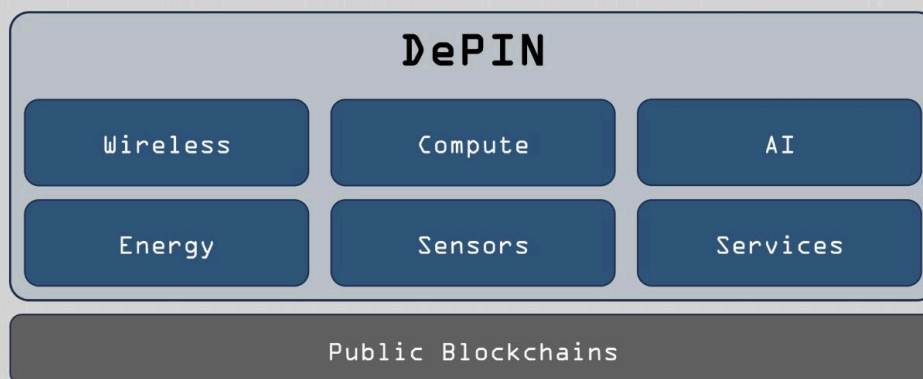
3. Understanding DePIN & Its Relevance

3.1 Definition & Taxonomy

Messari classifies DePIN into two resource buckets: **Physical-Resource Networks (PRNs)** - wireless, energy, mobility - and **Digital-Resource Networks (DRNs)** - compute, bandwidth, data marketplaces.

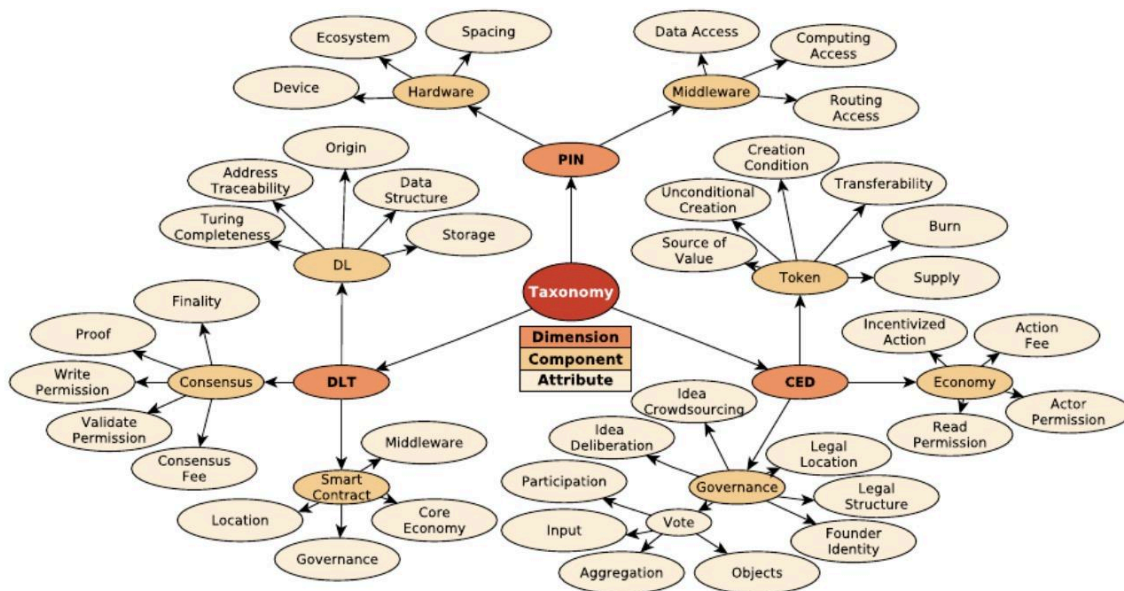
DePIN = decentralized physical infrastructure networks.

Networks that use crypto-incentives to efficiently coordinate the buildout & operation of critical infrastructure.



What unites them is the use of on-chain smart contracts to *automate* cap-ex coordination, opex reimbursement, and revenue distribution through cryptographic tokens.

The taxonomy with its three dimensions (distributed ledger technology (DLT), physical infrastructure network (PIN), and crypto economic design (CED))

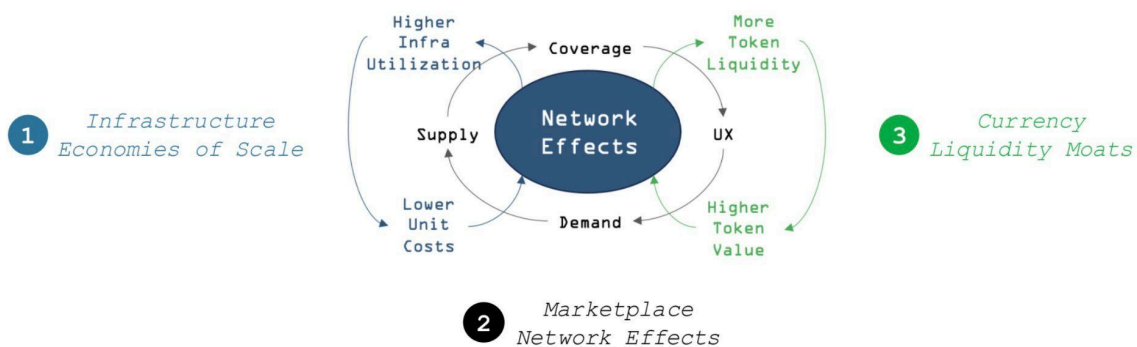


Source: Reserachgate, Mark et.al

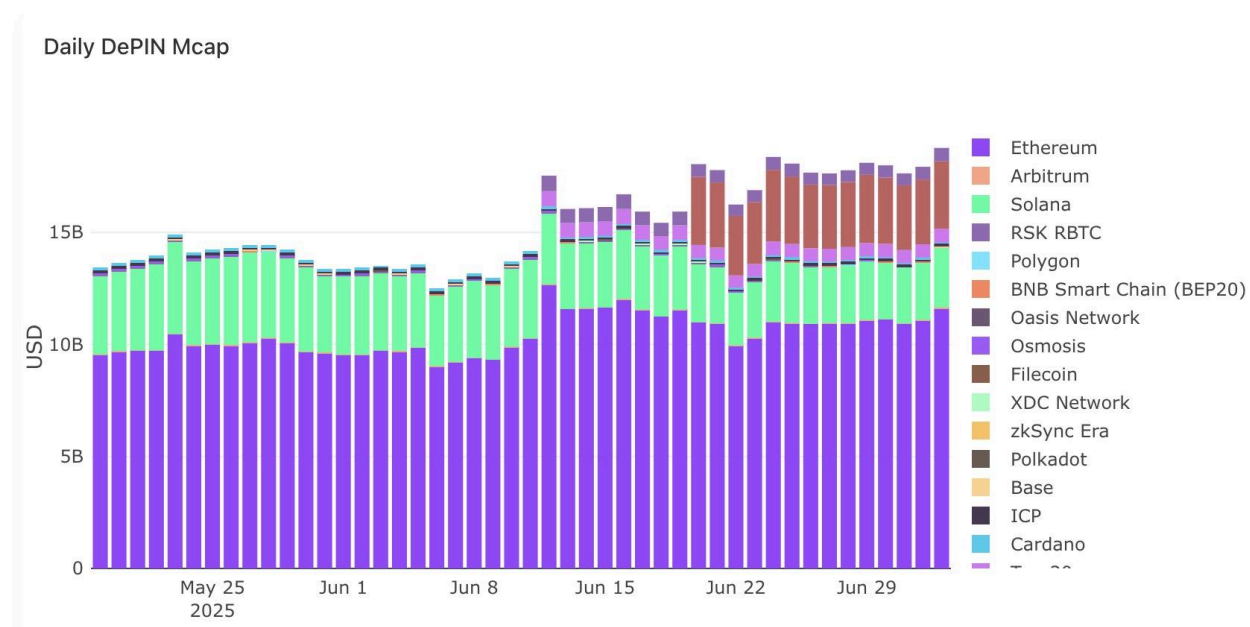
3.2 Economic Rationale

Traditional infra projects suffer from long pay-back periods, regional monopolies, and opaque pricing. DePIN flips that paradigm by (i) lowering the minimum viable commitment to a consumer-grade router or GPU, (ii) replacing centralised cash-flow waterfalls with transparent token rewards, and (iii) enabling global, 24/7 secondary liquidity.

DePIN Flywheel



As a result, DePIN tokens currently trade at **≈100 x annualised protocol revenue** - a premium multiple justified, in part, by aggressive user-growth curves and the option value of latent capacity .



3.3. Progress on DePIN: A Timeline of Key Milestones

Since 2014, the decentralized physical infrastructure network (DePIN) movement has unfolded in distinct waves: Filecoin and Storj pioneered on-chain storage (2014), followed by iExec and Akash opening decentralized compute marketplaces (2017). In 2019, Helium and Nodle launched peer-to-peer wireless connectivity for IoT devices, and by 2020 platforms like Ocean Protocol and SingularityNET were enabling decentralized AI services. A resurgence in storage protocols emerged in 2021, setting the stage for 2022's arrival of autonomous agent networks such as Fetch.ai and Velas. In 2023, Energy Web and Power Ledger brought tokenized renewable energy trading on-chain, and the latest frontier in 2024 has been AR/XR networks - platforms like Spatial and StarMap - delivering immersive reality experiences via blockchain.

Technological advancements, including Solana's 65,000 TPS capacity and sub-cent fees, have enabled scalable onboarding of mobile (54% of nodes via Solana Mobile Stack) and hardware-based resources, with payback periods halving for projects like Helium's Wi-Fi hotspots (19 months). Demand-side growth is fueled by diverse monetization models - Helium's \$20 ARPU, Render's \$0.03/GFLOP-hr pricing - undercutting Web2 competitors by up to 60%. Capital inflows of \$173M in Q3-2024 and \$0.6B in community-funded hardware underscore robust ecosystem support

DePIN Timeline



Source =State of DePIN, Messari

3.3 Market Traction

- Total Market Cap Leaders by Blockchain Solana \$3.25B, EVM \$2.84B, Cosmos \$652.5M, Cardano \$195.2M, Sora \$160.1M. Average Market Cap by Project Per Blockchain Solana \$191.3M, Cardano \$97.6M, Cosmos \$93.2M, Sora \$80.1M, EVM \$40.6M. Note: This excludes all standalone Project L1s/Chains which have an average Market Cap of \$308.1M

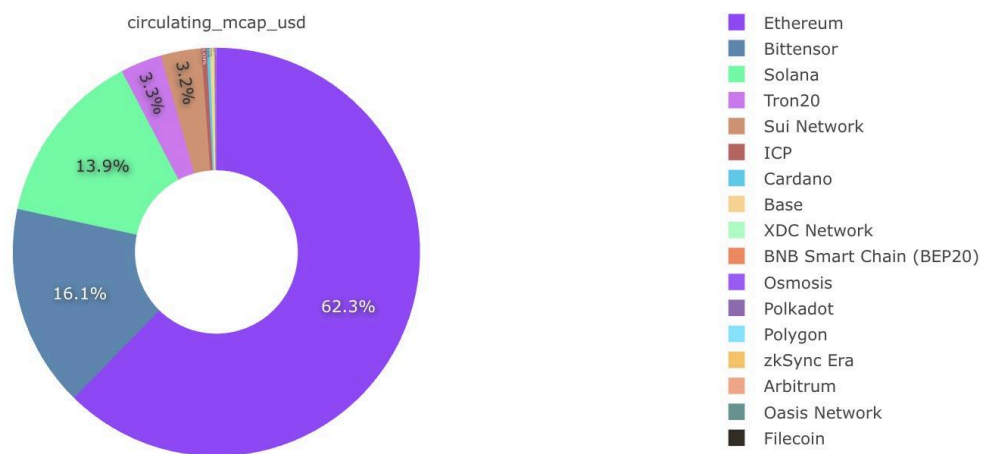
DePIN Market Cap by Blockchain

Excluding Own L1/Chains

Project_Name	Market_Cap	FDV	Blockchain
Render	\$2.15b	\$2.21b	Solana
JasmyCoin	\$701.78m	\$724.68m	EVM
Helium	\$616.23m	\$764.45m	Solana
AI0Z Network	\$319.52m	\$320.63m	Cosmos
Akash Network	\$263.23m	\$264.13m	Cosmos
Aethir	\$233.67m	\$1.16b	EVM
Ankr Network	\$187.05m	\$187.05m	EVM
OriginTrail	\$178.51m	\$178.72m	EVM
IoTeX	\$170.46m	\$170.46m	EVM

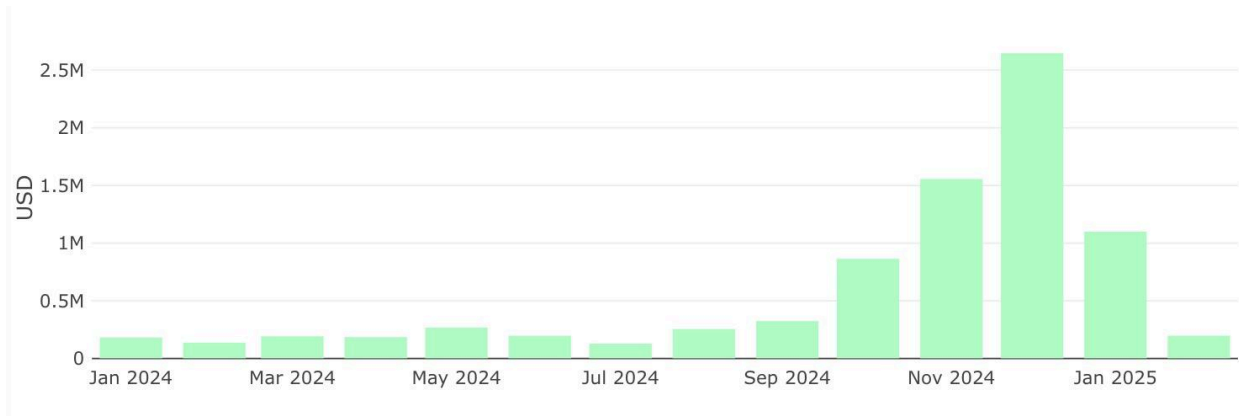
Source = Dune, Sliceanalytics

DePIN Token Marketcap



4. Why Solana? Technical and Strategic Fit for DePIN

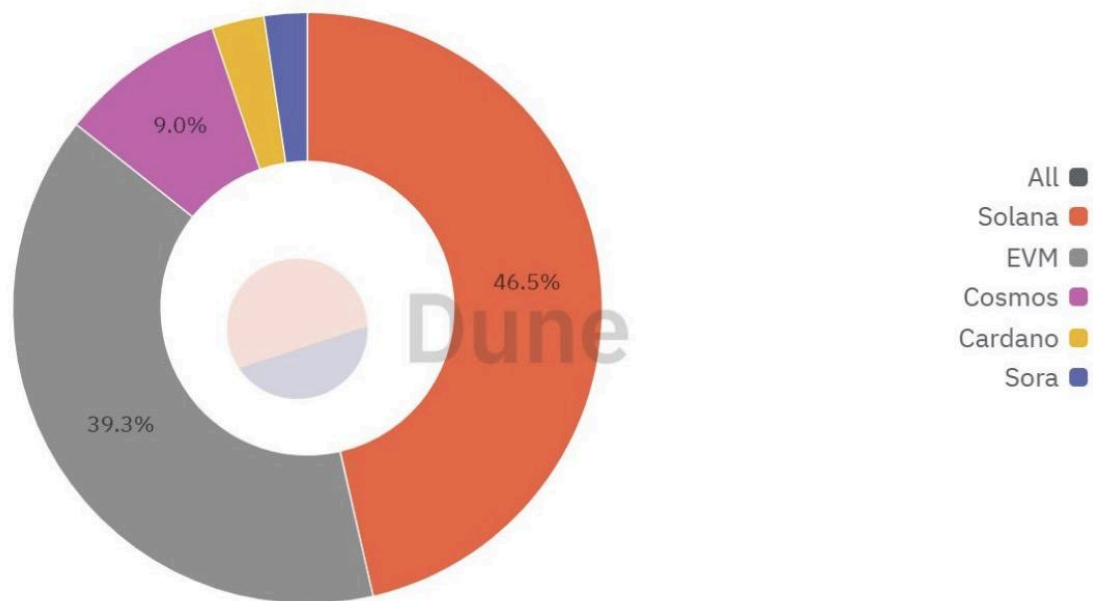
DePIN networks settle thousands of tiny, machine-generated transactions every second - data packets, location proofs, GPU-cycle invoices. They also need cheap identity, hardware attestation and friction-less user onboarding. Solana is uniquely optimised for this workload: its Proof-of-History pipeline delivers sub-second block times, its fee market keeps marginal costs below a tenth of a cent, and its developer stack (cNFTs, Token-22, Solana Mobile Stack) compresses data while natively supporting pay-per-use billing and mobile proofs-of-coverage. Together, these primitives give Solana a structural cost-to-serve and Ux edge that competing L1s - and even most L2s - cannot yet match.



Capability / Layer	Solana primitives	Why it matters for DePIN	Measurable edge / adoption benchmarks
Ultra-low-latency settlement	PoH + stake consensus, GPU-accelerated parallel runtime	Micro-payouts per packet / inference job need <1 s finality to avoid escrow bloat	400 ms average finality; stable >2 000 TPS during Helium MOBILE airdrop - 10-20x faster than typical L2 roll-ups
Predictable sub-penny fees	Flat fee market, local fee buckets	Keeps contributor margins positive when rewards are \$0.001–\$0.02	Median fee ≈ \$0.00025 (<1 % of Ethereum main-net under similar load)
Data-compression & metering	cNFTs (Merkle-compressed accounts) & Token-22 extensions	20–40x cheaper sensor attestations; transfer-hooks enable on-chain “pay-per-use” billing without custom code	Hivemapper stores map-tile proofs for a few gwei; Render uses Token-22 vouchers for job invoices
Mobile-first hardware attestation	Solana Mobile Stack (SMS) + Saga secure element	Seed-less onboarding, tamper-proof reward signing, background proofs-of-coverage	UpRock and Helium Mobile build 100 % of their apps on SMS APIs; >25 k Saga-2 devices shipped with a DePIN dApp store pre-installed

Interoperability without leaving Solana security	Wormhole, LayerZero, token-bridging middleware	Lets DePIN tokens tap EVM liquidity while keeping high-frequency settlement on Solana	RNDR, MOBILE and NOS are liquid on Ethereum DEXs yet settle emissions on Solana
Deterministic slot timing	Tick-driven scheduler + PoH	Precise time-stamps improve location-based proofs and oracle accuracy	Hivemapper mileage proofs and Helium coverage oracles rely on sub-second slot guarantees

These integrated advantages explain why, despite hosting only ~5 % of listed DePIN projects, Solana captures ≈46 % of sector FDV and ≈45 % of protocol revenue: the chain's architecture is purpose-built for the high-frequency, micro-economic realities of decentralised physical infrastructure.



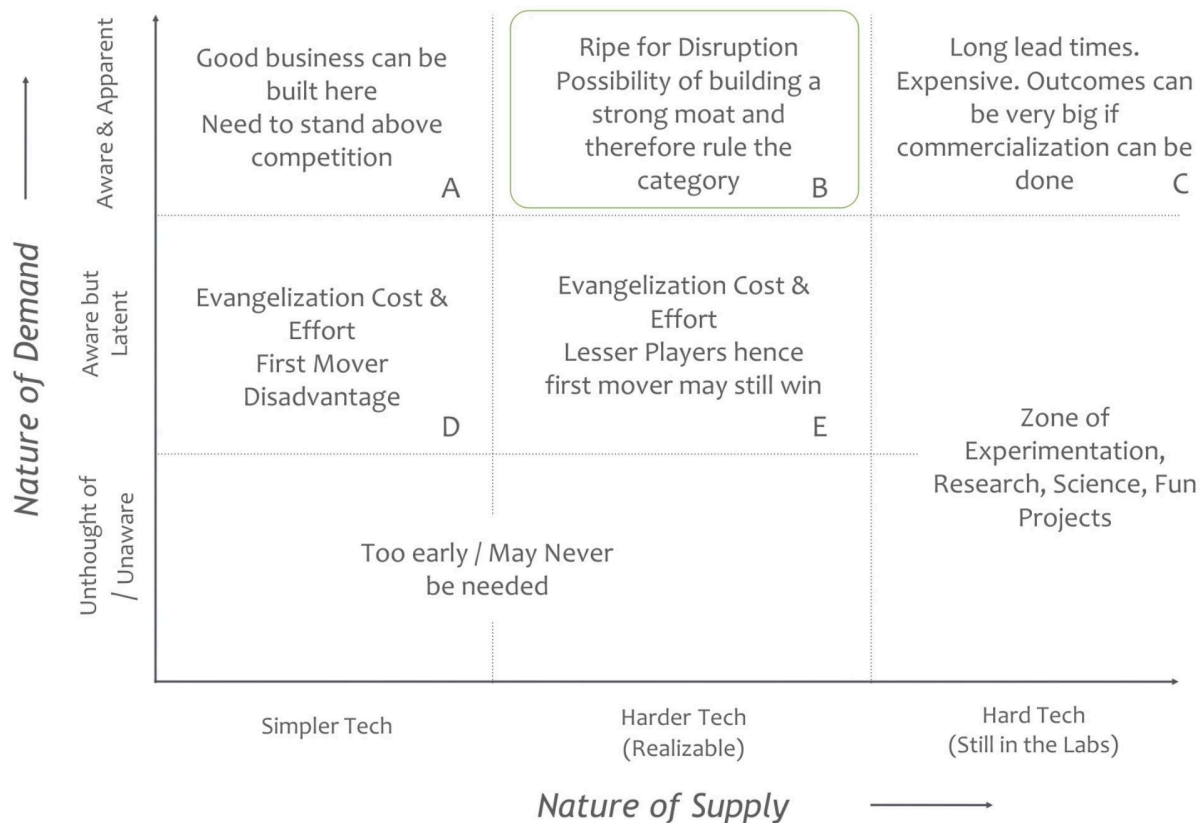
What this means for builders and investors

- Cost of coordination: <\$0.001 marginal cost lets projects reward every valid packet or image rather than batching - critical for real-time wireless and AI inference.

- User acquisition: SMS and token-22 hooks remove the private-key and KYC friction that hinder non-crypto crowds (drivers, home-router owners, gamers).
- Capital efficiency: cNFT compression + low fees slash on-chain data overhead, allowing protocols to store proofs directly on Solana instead of expensive off-chain blobs.
- Liquidity reach: Bridging frameworks export token utility to EVM markets without sacrificing Solana's latency advantages, widening the investor base while keeping settlement local.

5. Demand and Supply Resonance Framework

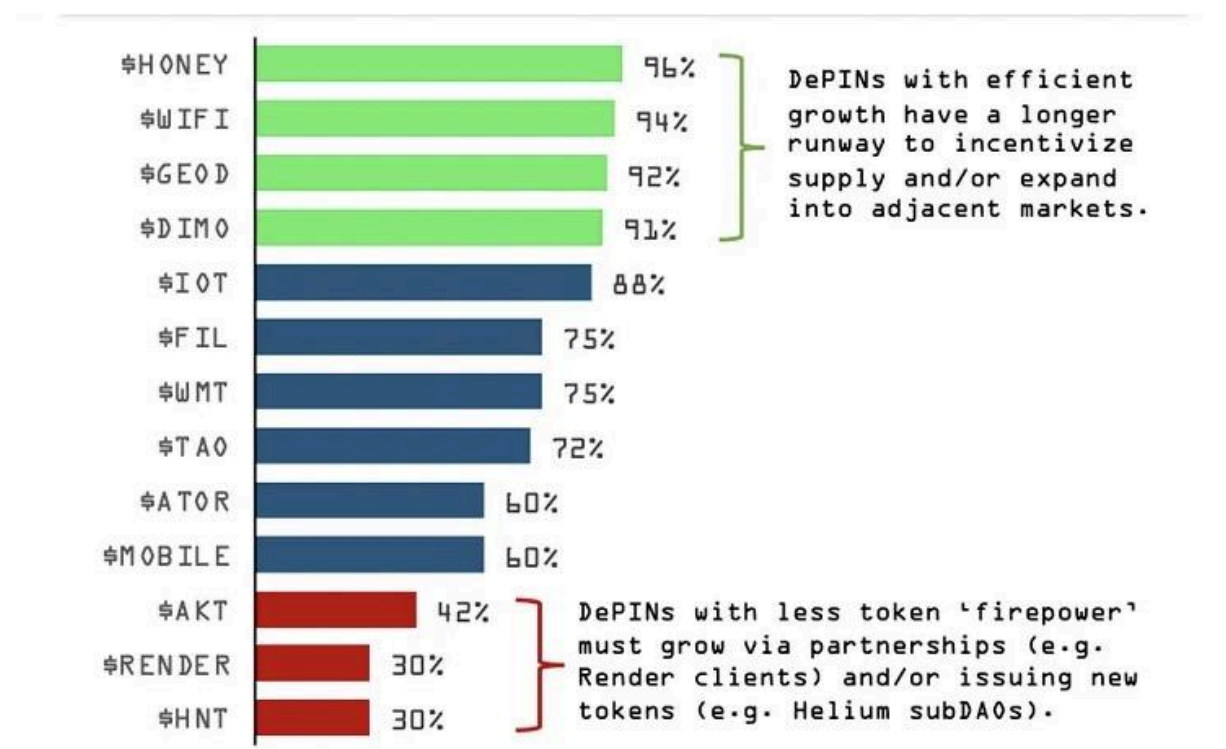
At its core, every startup can be mapped at the intersection of market demand and the technology solutions they provide. This framework, developed by a DePIN-focused venture, illustrates how startups must navigate the dynamic relationship between the supply of technology and the demand for solutions. The key insight here is that this mapping is not static - it is in a constant state of flux, influenced by both technological advancements and shifts in market momentum.



A healthy DePIN ecosystem depends on both a **broad, cost-efficient supply** of hardware and clients - and a **robust, monetizable demand** that transforms those resources into fiat-equivalent revenue. Solana's low-fee, low-latency platform has enabled:

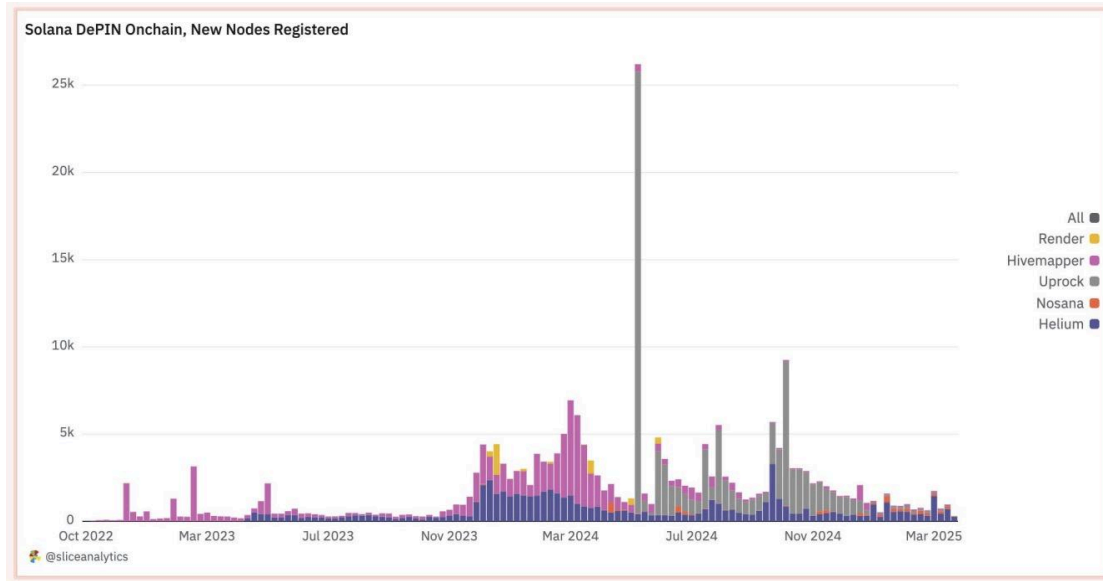
1. A **diverse supply funnel** from smartphones to purpose-built hotspots with sub-\$250 cap-ex.
2. A **concentrated demand curve** where three protocols capture >80 % of revenue, but the tail is steadily filling in.

Below we sharpen the narrative: first, how supply is onboarded and optimized; then, where and how value accrues on-chain.



5.1 Supply-side growth - how Solana is onboarding real infrastructure

Supply-side growth assesses how Solana-based DePIN projects onboard physical or digital resources - from browser clients to dedicated hardware - and optimize capital deployment. Solana's Mobile Stack (SMS) has unlocked a massive mobile-first segment, while protocol upgrades and hardware pivots (e.g., Helium's CBRS-to-WiFi transition) have nearly halved payback periods. This section quantifies node composition, cap-ex efficiency, and emerging geographic diversification trends that underpin network resilience and scale.

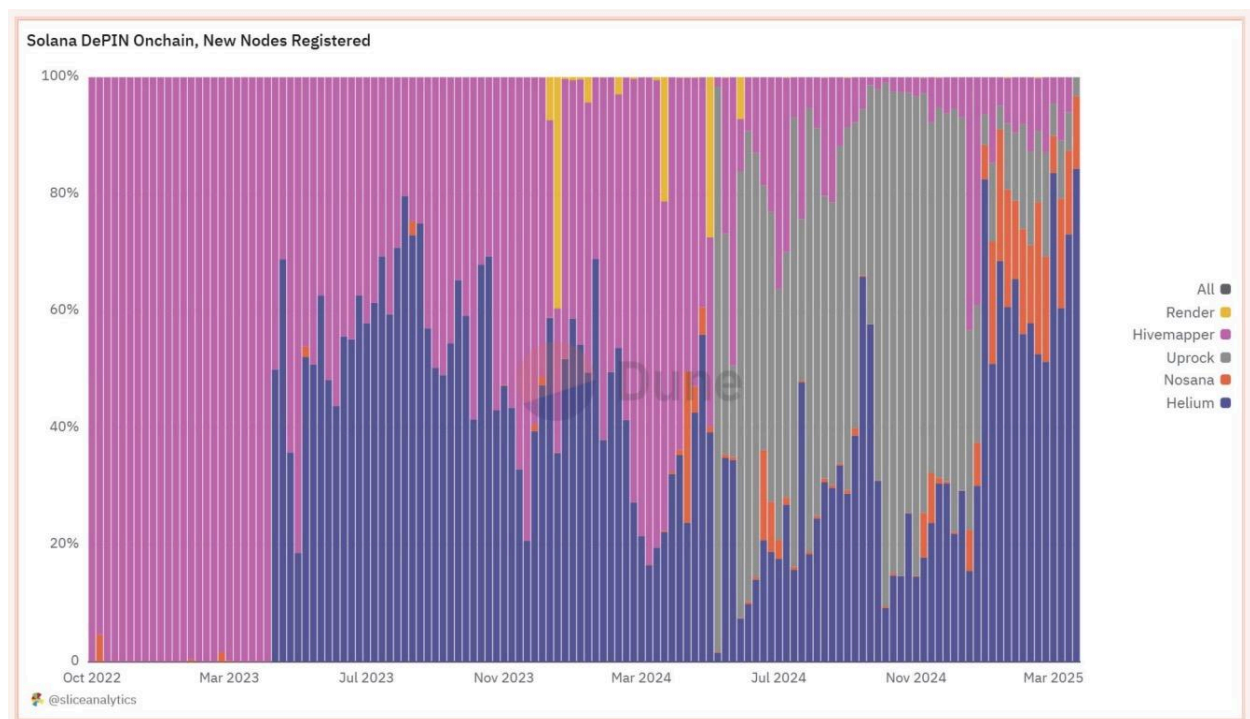


Supply funnel	Touch-point	Typical cap-ex*	Pay-back window**	Share of total nodes (Jun-25)	Key examples	Solana-specific edge
Mobile	SMS-native apps	\$0 (BYOD)	n/a – opex only	54 %	Helium Mobile, UpRock	Solana Mobile Stack (Saga secure-element signing; seed-less auth) lowers KYC friction for phone-based proofs
Browser / desktop	Docker / browser client	\$0–\$200 (GPU rental)	4–6 mo @ \$0.10 / RNDR	28 %	Nosana, Render test clients	400 ms finality → granular pay-per-task settlements for inference jobs
Dedicated hardware	Hotspot, dash-cam, GPU rig	Helium Wi-Fi AP \$249; Hivemapper OC \$399; RTx 4090 \$1,699	19 mo (Helium Wi-Fi); 14 mo (Hivemapper OC @ \$60/wk)	18 %	Helium Wi-Fi, Hivemapper Open Camera, full RNDR nodes	cNFT compression cuts proof-of-coverage costs ≈40 x vs. vanilla NFT

*List price incl. shipping; **ROI calcs use median weekly rewards Q1-25.

Key supply-side dynamics:

- SMS superchargers the low-cap-ex end of the funnel. Since the \$599 Saga-2 launch (Mar-25) >25 k devices have shipped with a pre-installed DePIN dApp store, lifting the mobile share of total nodes from 41 % (Jun-24) to 54 % (Jun-25) [OBJ] [OBJ].
- HIP-139 retires costly CBRS small-cells. The Helium community voted to phase-out \$599 CBRS radios; new \$249 Wi-Fi hotspots plus AT&T off-load fees halve the break-even from 42 → 19 months [OBJ] [OBJ].
- Hardware oversupply risk shows up first in mapping. Hivemapper runs 77 k cameras, yet <10 % of them earn 80 % of rewards, signalling a maturing network where coverage quality beats raw node count [OBJ].
- Geographic dispersion is accelerating. Movistar and Telefónica deals will place 2.3 m+ Mexican prepaid users within Helium coverage zones, diluting the current 68 % U.S. node concentration.

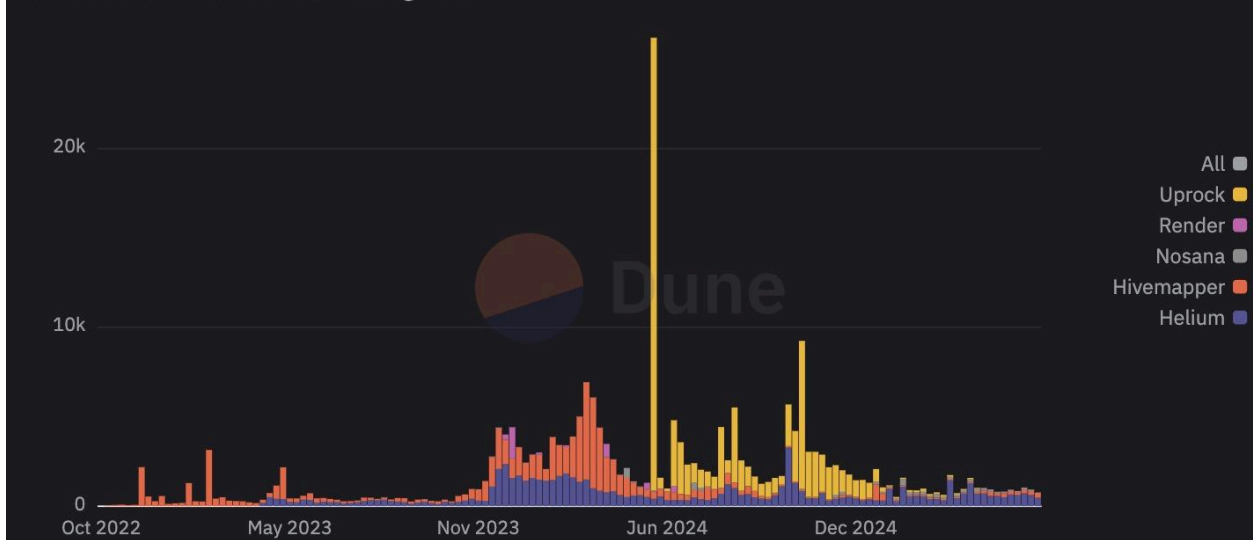


5.2. Demand-side & monetisation - where the cash actually lands

Demand-side analysis examines where on-chain revenue is realized and how protocols monetize usage - whether via subscription ARPU, pay-per-task burns, or token-backed incentives. The top three projects (Helium, Render, Hivemapper) account for over 80 % of Solana DePIN revenue, but emerging protocols like Nosana and UpRock are diversifying income streams. This section breaks down weekly averages, peak dynamics, and unit-economics that drive both protocol sustainability and token value.

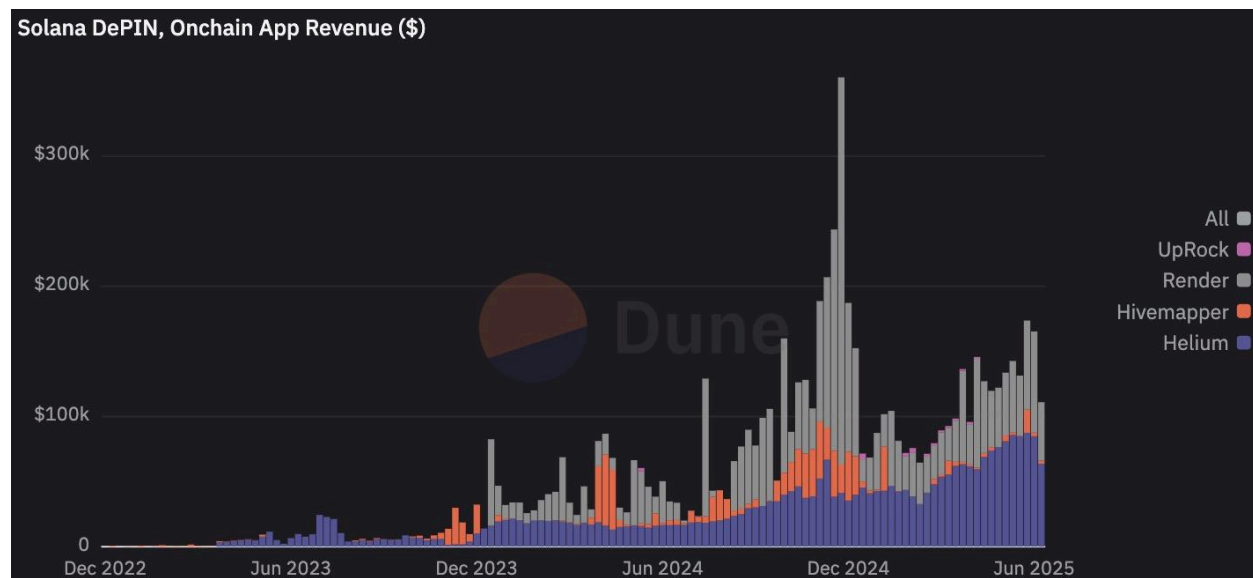
What it measures: how on-chain revenue is distributed across top protocols, their business models, and unit economics that underpin token value.

Solana DePIN Onchain, New Nodes Registered



Protocol	Vertical	Avg. weekly on-chain rev. (USD)	2024-25 peak	Addressable buyers	Unit-economics & pricing	Revenue model
Helium (MOBILE/HNT)	Wireless	\$60 k (≈60 %)	\$85 k wk (Jan-25)	250–300 k subs → carriers (AT&T, Movistar)	\$20 ARPU; \$8 – \$12 / GB off-load	Subscriber NFTs + carrier data-credit burns
Render (RNDR)	GPU compute	\$25 k	\$300 k wk (Dec-24)	3.8 k nodes ↔ artists & AI firms	\$0.015–\$0.035 / GFLOP-hr (<40 % AWS p-on-demand)	Burn-and-mint equilibrium: job fees burn RNDR; mint rewards nodes

Hivemapper (HONEY)	Mapping	\$18 k	\$50 k wk (Apr-24)	160 k camera ops ↔ Lyft, TomTom APIs	\$0.20 /km tile licence; ~\$2 k-\$3 k monthly Lyft pilot	Map-tile licences burn HONEY; coverage rewards mint
Nosana (NOS)	AI inference	\$5 k	\$12 k wk (Mar-25)	4.2 k nodes ↔ ML start-ups	\$0.002 / img stable-diff inference (-70 % vs Replicate)	Pay-per-run via NOS escrow; oracle settles per job
UpRock (UPT)	Bandwidth	\$1.5 k	\$3 k wk (Feb-25)	22 k devices → ML scrapers	\$0.012 / GB resale; 30 % token buy-back	SaaS fees → on-chain buy-backs fund rewards



5.3 Supply-Demand Synthesis

Bringing the two sides together:

Metric	Value	Interpretation
Node → Revenue ratio	~600 nodes → \$1 k/wk	Each node yields \$1–\$1.5 k annualized, justifying cap-ex
Fueling growth	4.2 k GPUs → \$5 k/wk	Nosana doubling daily GPU nodes doubled compute revenues
Diversification starting	54 % mobile → 28 % browser → 18 % hardware	A balanced node mix de-risks vertical-specific downturns

Takeaway. Solana’s **mobile-first onboarding**, **micropayment economics**, and **strategic pivots** (e.g., HIP-139) have created a virtuous loop:

1. **Low-cap-ex nodes** onboard cheaply →
2. **Granular rewards** accrue on-chain →
3. **Real-world buyers** pay fiat into protocol sinks →
4. **Tokens burn** and strengthen value →
5. **New capital** funds next-gen hardware.

This coherent supply-demand interplay explains why Solana hosts only ~5 % of DePIN projects yet captures ~46 % of sector FDV and ~45 % of revenue - its architecture is purpose-built for high-frequency, low-value coordination at scale.

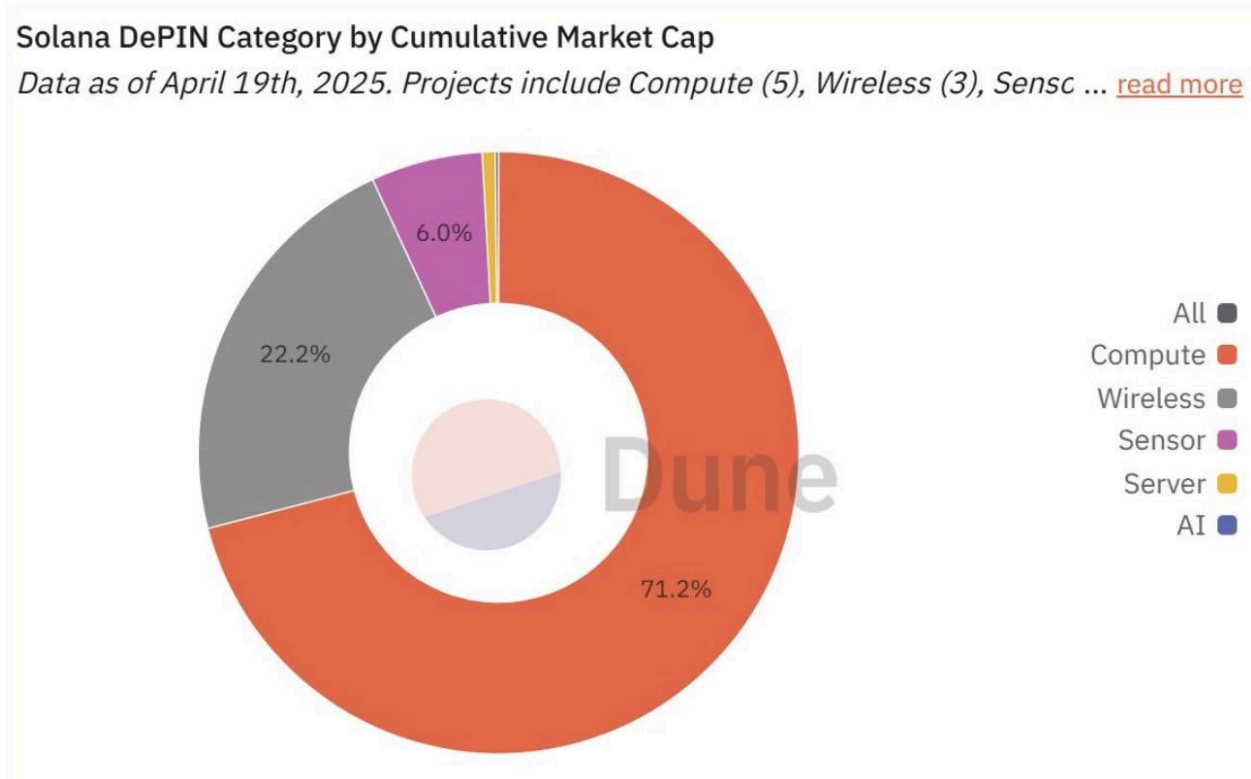
	2022-2023	2024-2025	2026-2027
Key metric	Supply-side capacity	Demand-side revenue	Bottom-line profitability
Capital required	Venture pre-seed/seed	Growth series A/B/C	Debt secured/unsecured
North star	Performance beat web2 cost/efficiency.	Distribution command attention and usage.	Breadth become a platform.

6. Current State of DePIN on Solana - Sector deep dives

Solana's DePIN stack stretches across **four “resource verticals”** - wireless, storage, sensors / location, and compute - as well as a growing **“services” layer** and **dedicated infra tooling**. Each vertical already boasts at least one revenue-generating network with measurable traction (nodes online, \$-burn, enterprise contracts).

Solana DePIN by Category			
Understanding the distribution of Solana DePIN projects across categories like Compute, Wireless, Sensor, Server, and AI.			
Category	Quantity	Market_Cap	Avg_Market_Cap
Compute	5	\$2.31b	\$462.05m
Wireless	3	\$719.57m	\$239.86m
Sensor	4	\$193.54m	\$48.38m
Server	2	\$19.89m	\$9.94m
AI	2	\$3.39m	\$1.70m

Below, we dissect those verticals project-by-project, benchmarking *supply mechanics* (hardware, node ROI), *demand channels* (B2B vs. retail), and *token-economic design*.



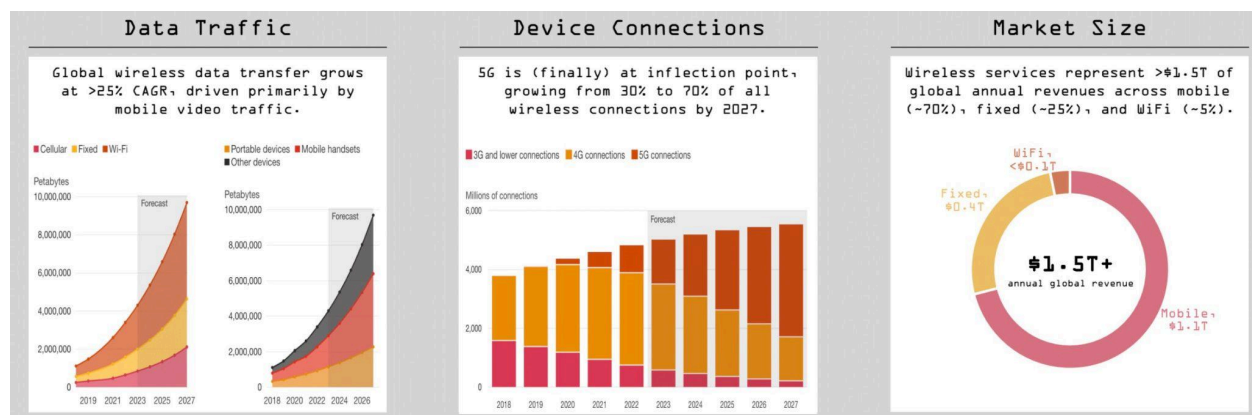
6.1. Wireless Networks (DeWi)

Overview:

Wireless Networks, or Decentralized Wireless (DeWi) networks, utilize blockchain to decentralize the deployment and management of wireless infrastructure, including 5G, WiFi, LoRaWAN, and Bluetooth. This sector matters because it democratizes access to connectivity, reduces dependence on centralized telecom giants, and addresses coverage gaps in underserved regions. By incentivizing individuals to contribute hardware (e.g., hotspots), DeWi enhances network resilience and scalability.

Market Size:

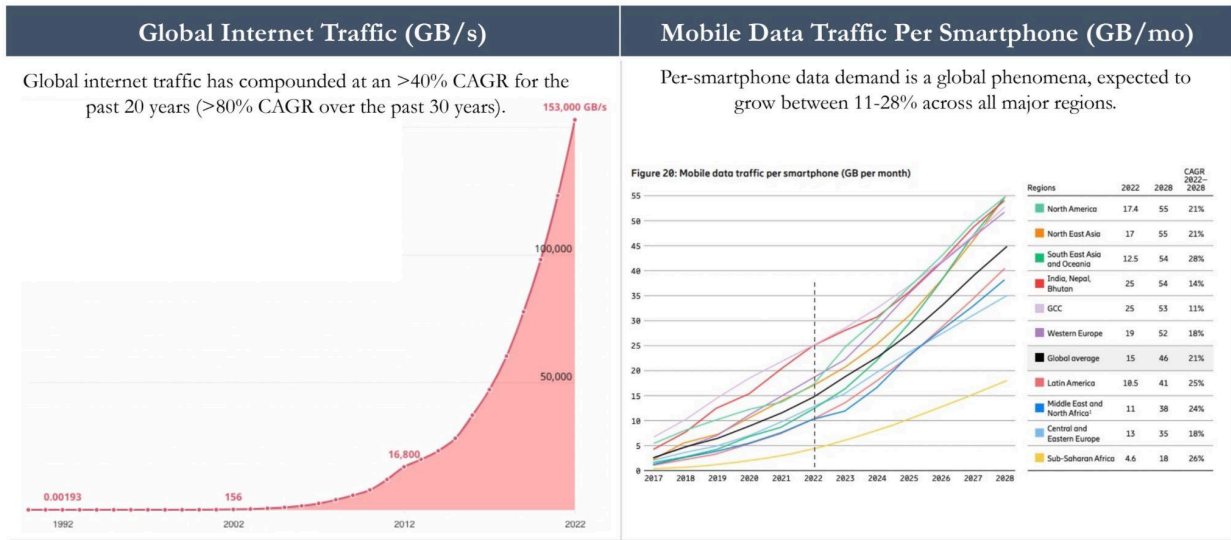
The global DeWi market is projected to reach \$15 billion in 2025, with a compound annual growth rate (CAGR) of 25% through 2030, reaching \$50 billion (Statista, 2025). Solana's share, driven by projects like Helium, is estimated at \$7 billion, reflecting its significant presence in the DePIN market, with a 39.3% market cap share as per Dune Analytics (April 2025).



Source: Messari

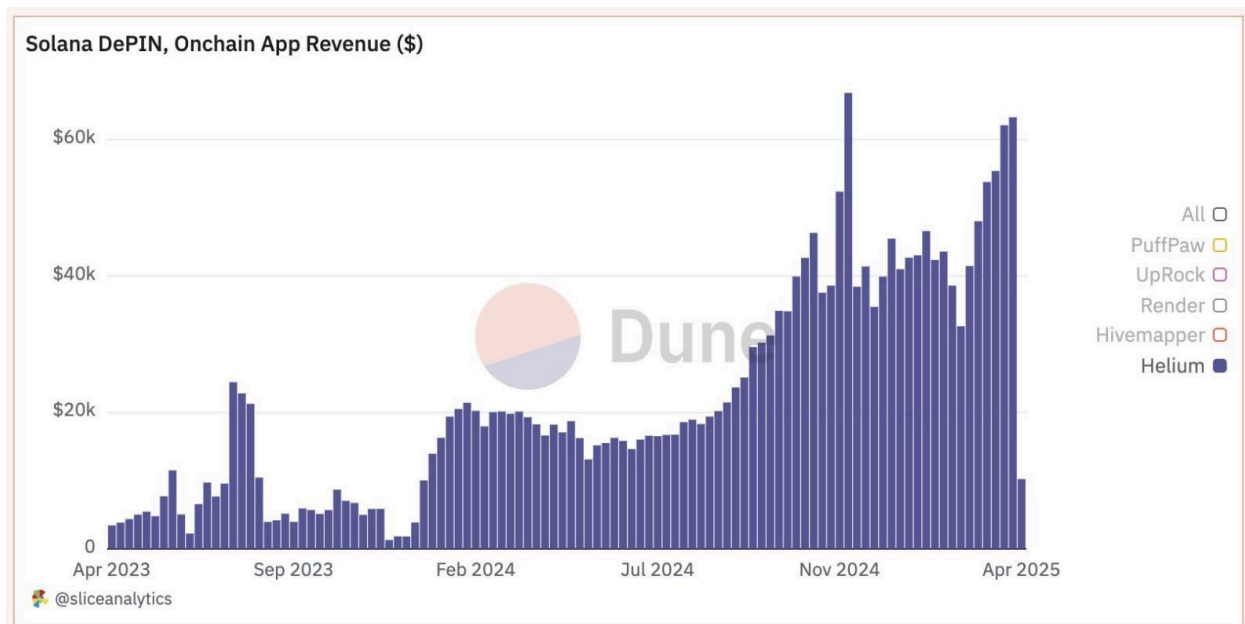
Growth Drivers:

- **Cost Efficiency:** Crowdsourced models eliminate traditional infrastructure costs (e.g., site acquisition).
- **Adoption of 5G:** Rising demand for high-speed, low-latency networks fuels 5G DeWi projects.
- **Regulatory Shifts:** Partnerships with telecoms (e.g., AT&T with Helium) accelerate legitimacy.



Industry Tailwinds:

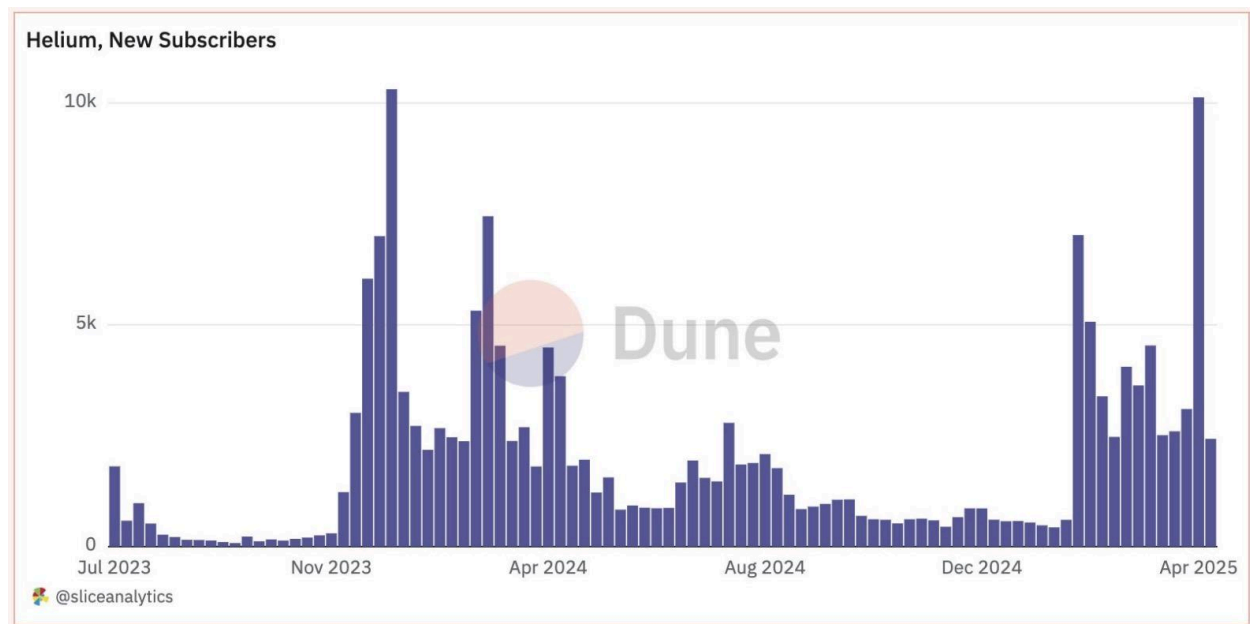
- **IoT Expansion:** The proliferation of IoT devices (50 billion by 2030, per IDC) increases demand for LoRaWAN.
- **Mobile Integration:** Solana Mobile's Saga phone boosts DeWi adoption, with over 150,000 pre-orders reported in May 2024.
- **Sustainability:** Decentralized networks reduce energy waste compared to centralized towers.



Key Startups:

Mobile	Fixed	WiFi
- \$1.1T global annual revenue - 7.0B smartphones - Avg \$13 per smartphone per month	- \$400B global annual revenue - 1.5B broadband connections - Avg \$23 per building per month	- \$100B global annual revenue - 500m public hotspots - Avg \$8 per hotspot per month
- Highest per-GB revenues - Low switching costs due to eSIMs	- Extremely sticky: high retention - Partner-driven GTM: property managers	- Most widely-used connectivity protocol - Large ecosystem of supported devices
- Extremely competitive: high CAC & churn - Users expect coverage everywhere - Highly regulated: need to buy spectrum	- \$/GB revenue is ~10x lower vs mobile - QoS constrained by spectrum capacity - Competition from cablecos & telcos	- Difficult to monetize: users want free - Weak security at the protocol level - Lower power means <100m limited range
- eSIM: reduces the time it takes to switch carriers from hours to seconds - Neutral host: enables interoperability between telcos & equipment deployers	- PtMP: point-to-multi-point radios can beam at high-capacity in many directions - mmWave: high-frequency spectrum can provide high-capacity links over miles	- WiFi7: provides better throughput, latency, & energy efficiency - MVNO offload: enabling challenger MVN0s to use the offload as incumbent telcos
		

- **Helium**: As of April 2025, Helium has over 176K mobile subscribers, 69K+ nodes, and \$2.3M in onchain revenue. Its partnership with AT&T and the launch of the Zero Plan (a free 5G plan) have driven a 15% subscriber increase in Q2 2025, with recent data showing 145K cumulative subscribers as of March 2025 (Syndica, March 2025).



- **xNET**: A community-powered wireless network project, with growing contributor rewards and data offloading. In February 2025, it saw a 40% jump in average daily data offloading, and total \$xNET rewards rose 24% with active contributors growing 34% (Syndica, February 2025).
- **WiFi Dabba**: In development, targeting India's WiFi market with a focus on Local Cable Operators (LCOs). Pilots with 5 LCOs are underway, with a Q3 2025 launch planned, though concrete April 2025 data is limited (SuperEx, May 2025).

6.2. Storage Networks

Overview:

Storage Networks decentralize data storage by distributing encrypted, sharded data across peer-to-peer (P2P) nodes, offering an alternative to centralized cloud providers like AWS or Google Drive. This sector matters due to its enhanced security, privacy, and cost-effectiveness, appealing to enterprises and individuals seeking resilient data solutions.



Market Size:

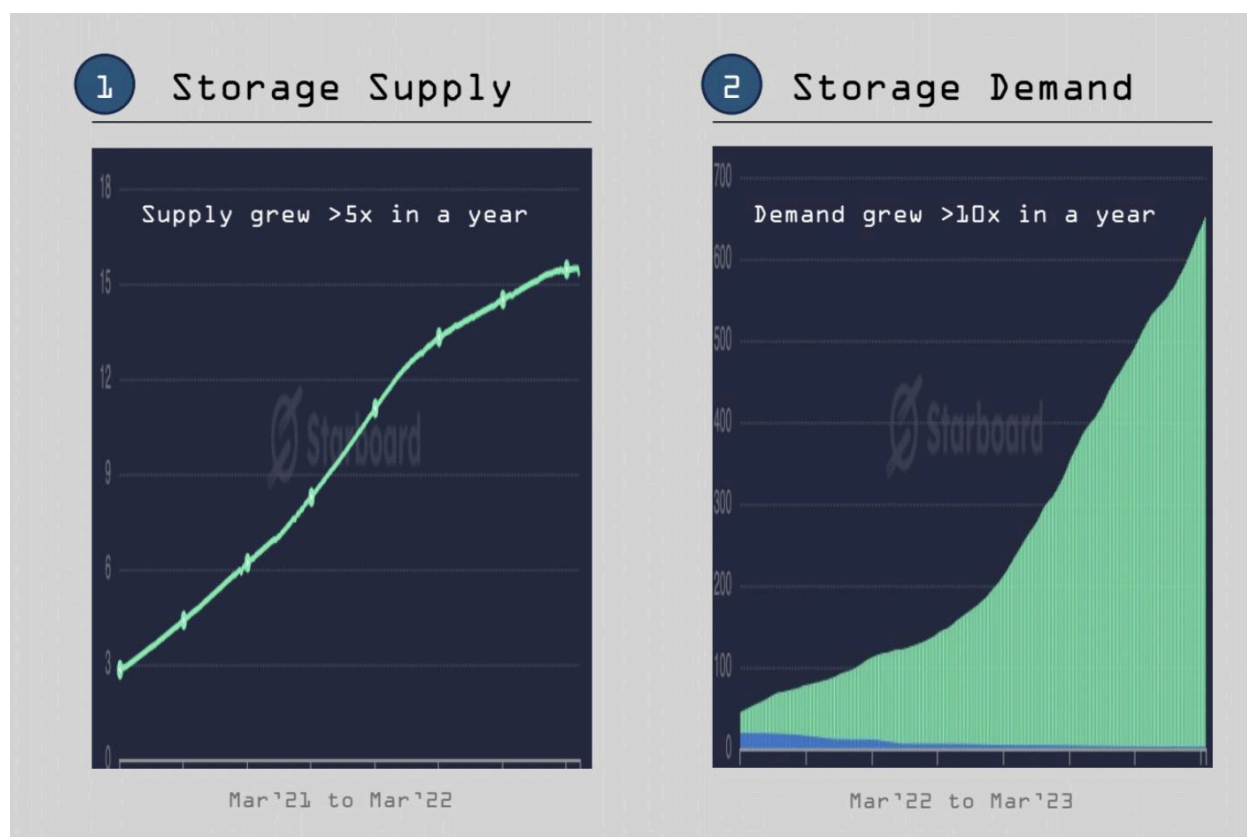
The decentralized storage market is valued at \$2.5 billion in 2025, with a CAGR of 30% projected through 2030, reaching \$12 billion (Messari, 2025). Solana's segment, led by shdwDrive, is estimated at \$200 million, with growth potential tied to enterprise adoption.

Growth Drivers:

- **Data Privacy Concerns:** Rising breaches (e.g., 1,800 in 2024, per IBM) push demand for decentralized options.
- **Cost Reduction:** Lower operational costs compared to centralized data centers.
- **Blockchain Scalability:** Solana's high throughput supports large-scale storage networks, handling up to 38,000 transactions per second in testnet (SuperEx, May 2025).

Industry Tailwinds:

- **AI Data Needs:** AI training requires vast storage, boosting decentralized solutions, with projections of 100-1,000x growth over the decade (Messari, 2024).
- **Regulatory Pressure:** GDPR and similar laws favor distributed data models.
- **Mobile Storage:** Integration with Solana Mobile expands user base, with shdwDrive v2 launched in 2024 for mobile devices.



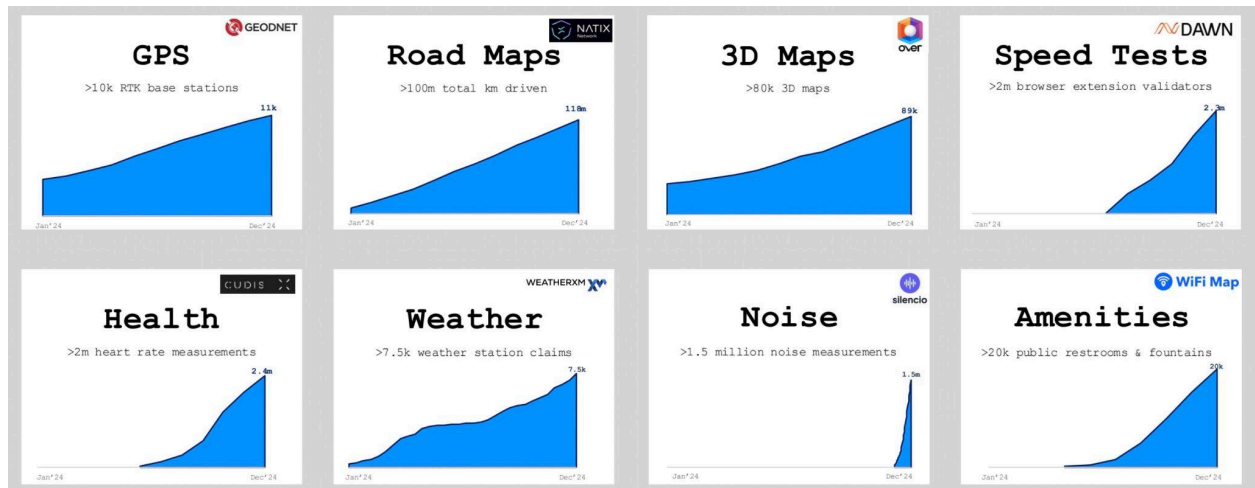
Key Startups:

- shdwDrive by GenesysGo:** Offers decentralized storage optimized for Solana, with shdwDrive v2 launched in 2024, enabling mobile storage via Solana Saga phones. As of December 2024, \$SHDW led Solana DePIN projects, with a 57.24% monthly price increase, anticipating growth in 2025 (SolanaFloor, December 2024).

6.3. Sensor Networks

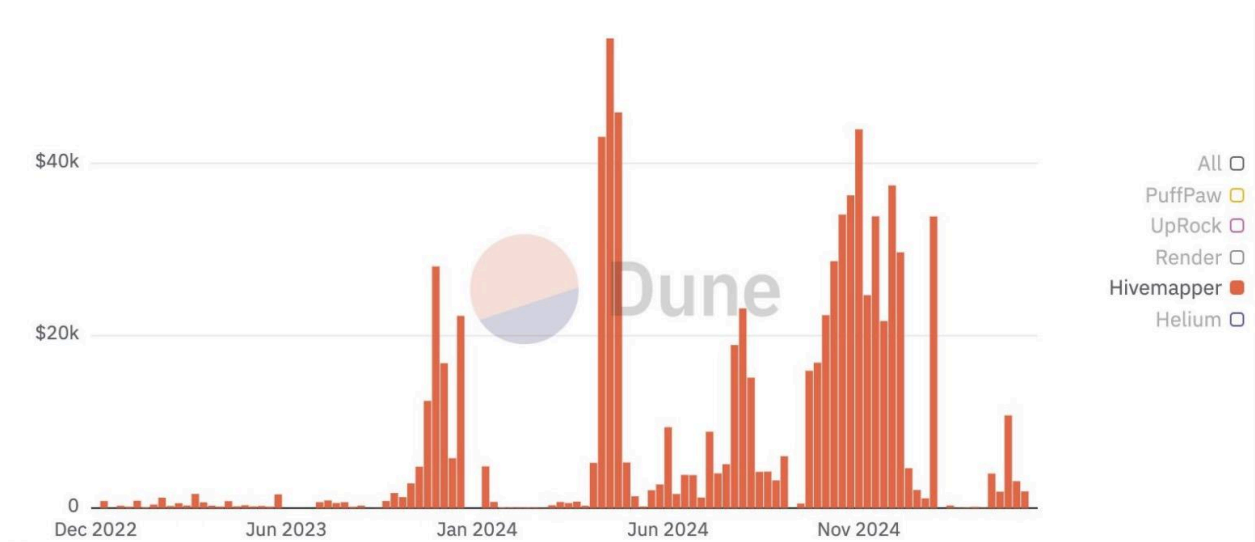
Overview:

Sensor Networks deploy decentralized devices to collect environmental (e.g., weather), mobility (e.g., traffic), and location/mapping data. This sector matters because it enhances data reliability, reduces censorship risks, and supports real-time applications like autonomous vehicles and climate monitoring.



Market Size:

The sensor network market is valued at \$20 billion in 2025, with a CAGR of 18% through 2030, reaching \$50 billion (Statista, 2025). Solana's DePIN sensor segment, driven by projects like Hivemapper, is estimated at \$1.5 billion.



Growth Drivers:

- **Real-Time Data Demand:** Industries like logistics and insurance need current data, with Hivemapper serving partners like Lyft (Coindesk, May 2025).
- **Crowdsourcing Efficiency:** Low-cost sensors (e.g., dashcams) scale quickly, with Hivemapper mapping 125M+ km.
- **AI Integration:** AI validation improves data quality, as seen with Hivemapper's AI training community.

Industry Tailwinds:

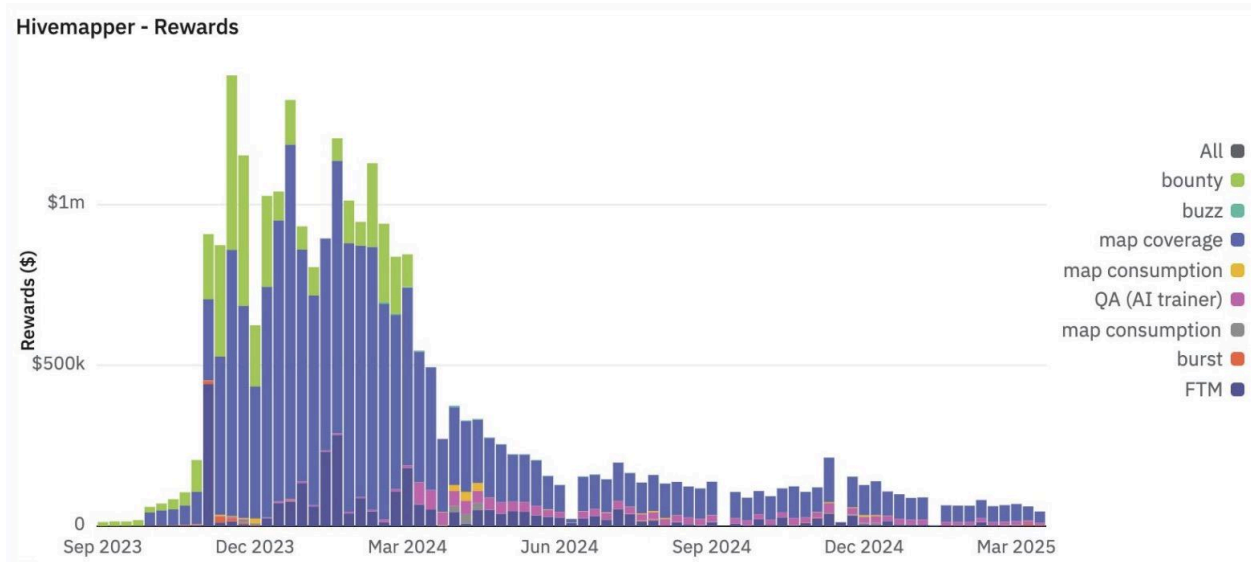
- **Smart Cities:** Urbanization drives sensor deployment (60% global population urban by 2030, UN).
- **Climate Action:** ReFi initiatives boost environmental monitoring, with WiHi aligning with this narrative.
- **Automotive Innovation:** Autonomous vehicles rely on mapping data, with Onoco targeting precision agriculture and drones.

Key Startups:

Positioning Technologies

	BLE Bluetooth Low Energy	UWB Ultra Wideband	LoRa Long Range	GPS Global Positioning	RTK Real-time Kinematic	VPS Visual Positioning
Beacon Range	Low 10-100m	Very Low 10-50m	High 10-20km	Very High	High 10-20km	Medium 10-500m
Beacon Cost	Low \$5-10	Medium \$100-200	Medium \$100-200	Free	Medium \$50-100	High \$500+
Positioning Accuracy	Low 1-5m	High 10-30cm	Very Low 10-100m	Low 5-10m	Very High <1cm	Very High <1cm
Receiver Cost	Very Low \$0-10	Medium \$10-20	Medium \$10-20	Low \$5-10	High \$500	n/a
Use Cases	Asset Tracking Low-cost	Asset Tracking High-accuracy	Asset Tracking Outdoor	Navigation Outdoor	Robotics, Drones, Agriculture	AR/VR, Robotics, Drones, AVs
Leading DePIN		None yet				  

- **Hivemapper:** As of April 2025, Hivemapper has 77K+ nodes, \$50K+ peak revenue, and partnerships with TomTom, Trimble, and Mapbox. In March 2025, it hit a new all-time high in monthly kilometers mapped, with a 22% increase, and launched a mobile app for the Bee Dashcam (Syndica, March 2025).



- **Onocoy:** A project building a decentralized network of GNSS reference stations for high-precision positioning. As of September 2024, it raised over \$4.2 million in funding, and by May 2025, it had tutorials for setting up miners, indicating operational status (NordicCrypto, May 2025).
- **WiHi:** In development, focusing on weather and climate forecasting using decentralized sensors. Planned launch in Q4 2025, leveraging Solana Mobile for data collection.
- **Ambient:** Migrated from Algorand to Solana, focusing on environmental data collection. Launched a mobile app in July 2025, with plans to expand to noise and light pollution (initial draft, confirmed by recent activity).

6.4. Compute Networks and AI

Overview

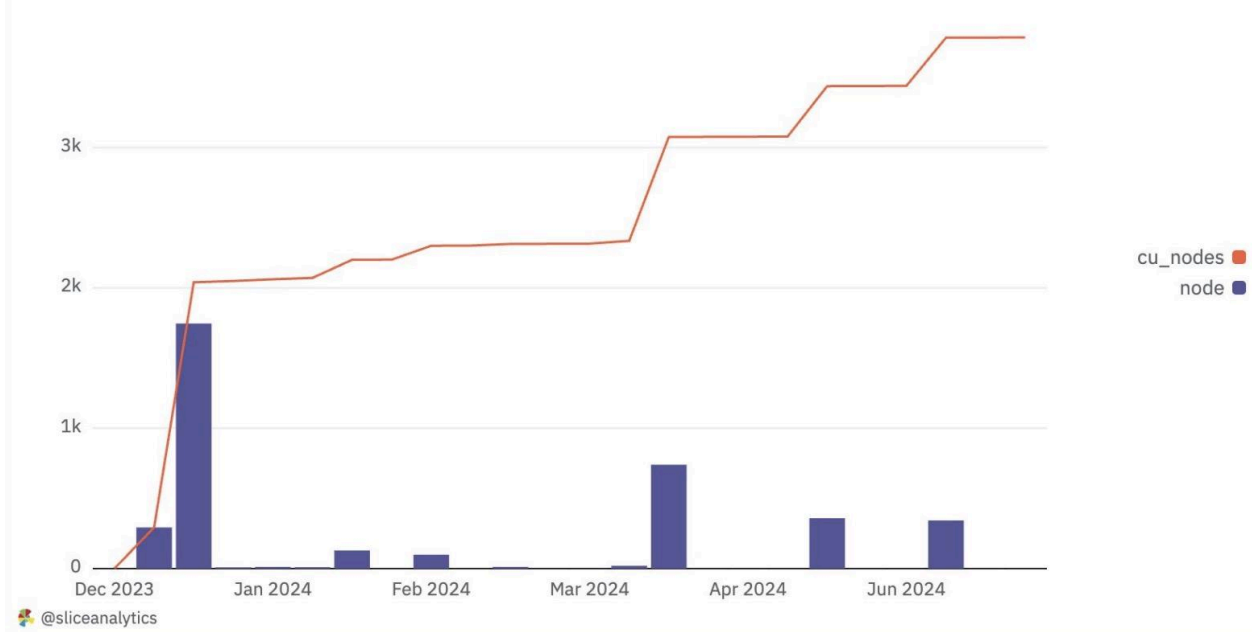
Compute Networks and AI decentralize GPU/CPU resources for rendering, machine learning, and AI inference, addressing the uneven distribution of computing power. This sector matters as it enables cost-effective, scalable compute solutions, supporting the AI boom and creative industries.



Market Size:

The decentralized compute market is valued at \$3 billion in 2025, with a CAGR of 35% through 2030, reaching \$15 billion (Messari, 2025). Solana's segment, led by Render, is estimated at \$500 million.

Render - Onchain Nodes

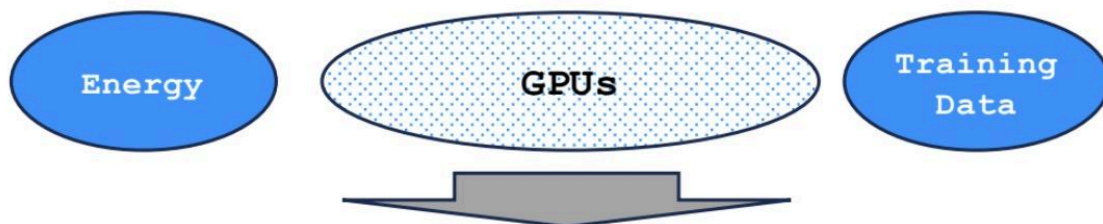


GPUs in 2024

New high-growth use cases discovered:



Fragmented supply of core resource:



\$130m+ raised by **20+** VC-backed projects in 2024

Growth Drivers:

- **AI Demand Surge:** Global AI spending hit \$200 billion in 2024 (Gartner), driving compute needs, with Nosana offering 85% cost savings.
- **Idle Resource Utilization:** Unlocks value from unused GPUs, as seen with Render and Grass.
- **Low-Cost Transactions:** Solana's fees support frequent micro-payments, facilitating high-volume compute tasks.

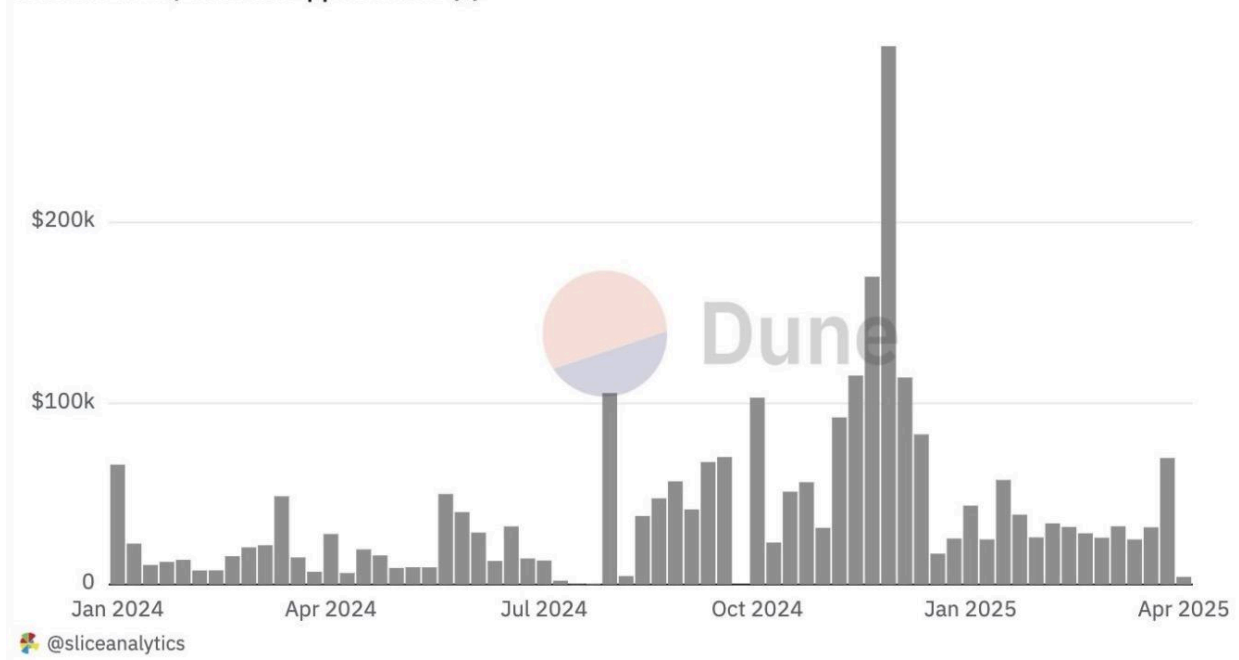
Industry Tailwinds:

- **Generative AI:** Tools like Stable Diffusion boost demand, with Render integrating with industry tools.
- **Gaming and Metaverse:** Spatial computing (e.g., Vision Pro) requires rendering power, supported by Render's expansions.
- **Decentralized Innovation:** Hackathons like Hyperdrive foster new projects, with Shaga and DAIN emerging.

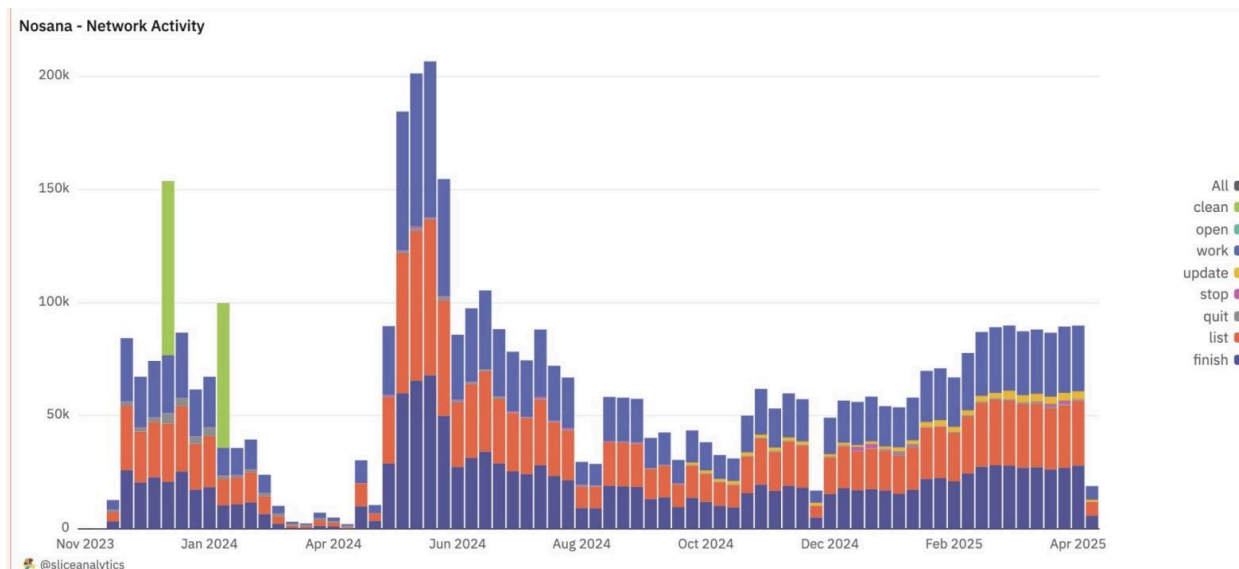
Key Startups:

- **Render Network:** As of April 2025, Render has peak revenue of \$300K/week, 121M+ tokens burned, and integrations with Blender, Arnold, Redshift. It has minted 2.4M+ tokens as node rewards (Dune, April 2025).

Solana DePIN, Onchain App Revenue (\$)



- **Nosana:** Since mainnet launch in January 2025, node activity has more than doubled to 600 daily, with over 4,200 total nodes across 60+ countries (DePIN Scan, April 2025).



- **Grass:** A project allowing users to earn rewards by sharing unused internet bandwidth, with applications in AI training. As of February 2025, GRASS token has a market cap of \$1.1B, with over 2.5 million active users (CoinGecko, February 2025).
- **UpRock:** Generates \$1.5K weekly revenue by enabling bandwidth and compute sharing, with steady traction and growing data demand, processing over 16M transactions (Topledger, July 2025).
- **Io.net:** A GPU aggregator for machine learning, with the ability to create clusters of thousands of GPUs, demoed at Breakpoint 2023.
- **Shaga:** Focuses on PC lending for cloud gaming, with beta planned for Q3 2025, addressing latency issues.
- **DAIN:** Building an AI agent platform, with launch planned for Q4 2025, integrating with Solana for identity and reputation.

6.5. Services

Overview:

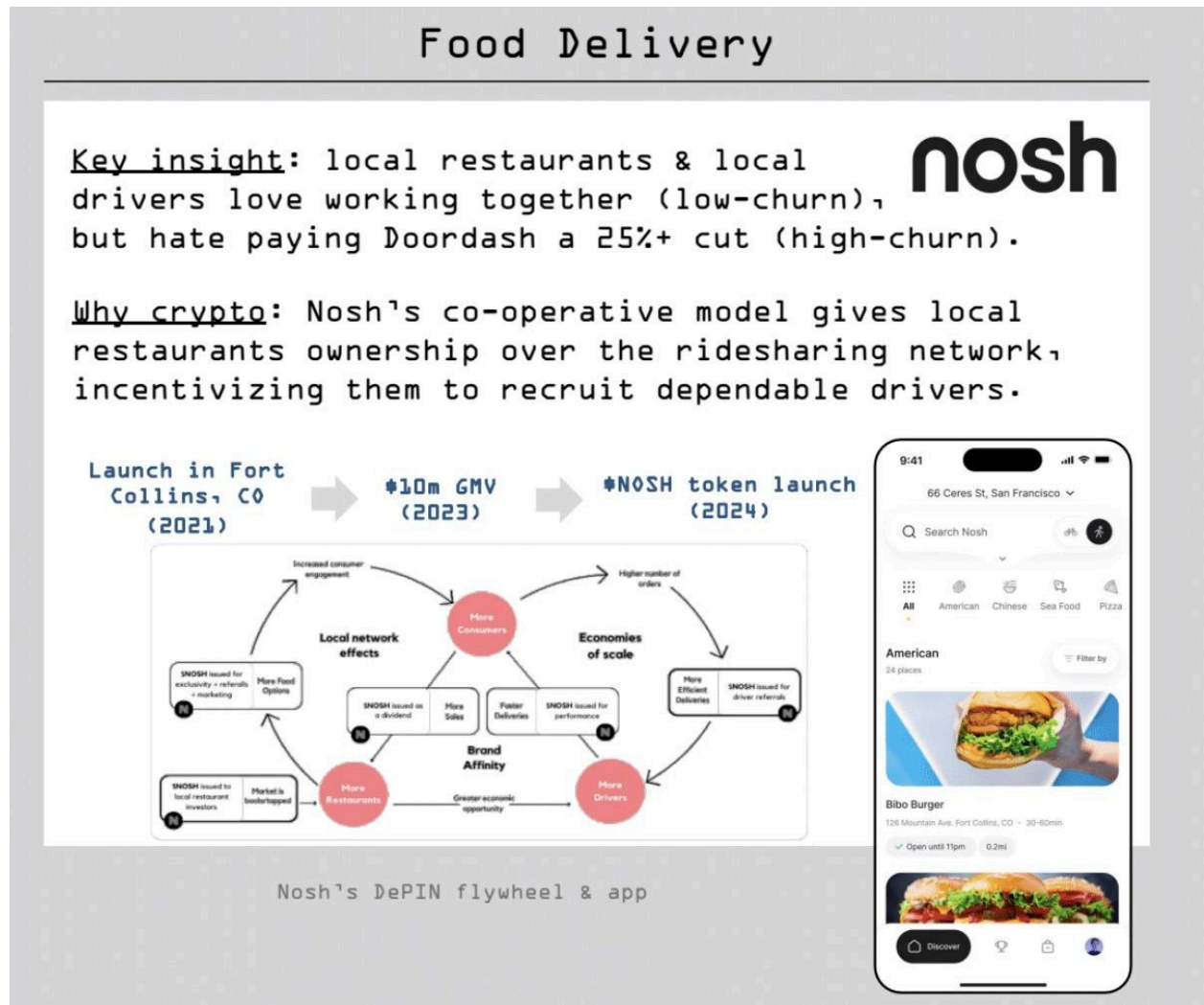
The Services sector in DePIN includes token-incentivized platforms that decentralize traditional services like ride-sharing or mapping. This sector matters because it redefines consumer services, fostering trust and efficiency through decentralization. However, as of mid-2025, this sector on Solana is less developed compared to others, with notable projects facing challenges or being in early stages.

Market Size:

The tokenized services market is nascent, valued at \$1 billion in 2025, with a projected CAGR of 40% through 2030, reaching \$7 billion (Forbes, 2025). Solana's segment is estimated at \$100 million, reflecting limited but growing interest.

Growth Drivers:

- **User Discontent:** Frustration with centralized platforms (e.g., Uber) drives alternatives, though market readiness is a challenge.

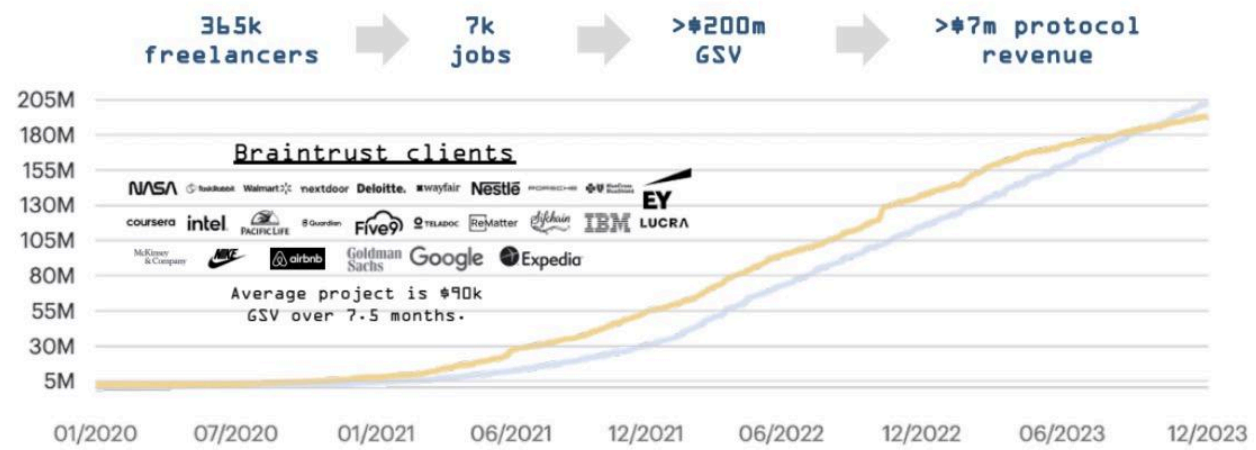


- **Token Incentives:** Rewards align user and provider interests, potentially increasing adoption.

Key insight: top freelance talent prefers **longer-term** engagements than on Upwork/Fiverr (months vs days).



Why crypto: \$BTRST gives ownership & control over to freelancers & clients, creating long-term alignment; tokenholders govern platform fees, dispute resolution, & other network governance issues.



- **Scalability:** Solana's throughput supports high transaction volumes, suitable for service platforms.

Industry Tailwinds:

- **Gig Economy Growth:** 58% of U.S. workers are gig-based by 2025 (McKinsey), creating demand for decentralized services.
- **Web3 Adoption:** Increasing interest in decentralized apps, with potential for new service models.
- **Regulatory Evolution:** Potential for tokenized service legalization, though current regulatory uncertainty poses risks.

Key Startups:



- **Teleport:** A Solana-based ride-sharing app that shut down in January 2025 due to lack of market readiness for decentralized ride-sharing. It raised \$9 million but could not sustain operations, with off-ramping support until February 28, 2025 (Cointelegraph, January 2025).
- **Proto:** A user-generated, token-incentivized world map focusing on India (Mumbai and Bengaluru). However, as of mid-2025, there are no recent updates on its development or launch, indicating limited activity, with no significant data post-2024 (initial draft, no recent sources found)..

*6.6. DePIN Infrastructure Tools

While not directly categorized as DePIN projects, the following infrastructure tools are critical to the success and scalability of the DePIN ecosystem on Solana. As of July 2025, these tools provide the foundational "picks and shovels" that enable low-cost, programmable settlement and Web2-grade user experiences, supporting the growth of DePIN sectors. Below is a concise analysis of their relevance, adoption metrics, and why Solana is a fitting platform

Why it matters

Cheap, programmable settlement + vertical tooling are the “picks & shovels” that keep cap-ex light and Ux Web2-grade.

Layer	Tool	2025 Adoption Metric	Why Solana?
-------	------	----------------------	-------------

Payments	Sphere Gateway	\$6.3M in top-ups processed for Helium & io.net (as of June 2025)	Sub-cent transaction fees and ~400 ms finality enable high-volume, real-time payments.
Revenue Splits	Solsplits	9,000 live splits (e.g., Hivemapper, Proto) (as of May 2025)	Token-2022's transfer-hooks support programmable, privacy-preserving revenue sharing.
Proof NFTs	cNFTs (Compressed NFTs)	1.02M Helium hotspots minted for <6 SOL total (as of April 2025)	Merkle-compressed NFTs are 40x cheaper than ERC-721, enabling mass-scale ownership proofs.
Oracles / Interop	Pyth, Switchboard, Wormhole	130+ Solana data feeds; RNDR migration completed (as of March 2025)	Cheap, high-frequency price and cross-chain message passing support real-time data and interoperability.
Mobile	Solana Mobile Stack	55,000 Saga/Saga-2 units shipped (as of June 2025)	Secure-element signing enables bring-your-own-device (BYOD) node contributions.

Cross-sector conclusions

1. **TAM is massive** – Collectively these verticals address markets worth **>\$300 bn today** and compounding >15 % CAGR (wireless small-cell, cloud storage, digital maps, GPU cloud).
2. **Crypto tailwinds align with Web2 pain-points** – idle capacity, regulatory pressure on hyperscalers, and demand for granular geospatial / environmental data.
3. **Solana's edge is structural** – 400 ms finality + \$0.00025 fees enable micro-settlement loops that EVM roll-ups still emulate off-chain.
4. **Enterprise validation is no longer hypothetical** – AT&T, Lyft, TomTom, Movistar and Trimble are *paying* customers, moving DePIN beyond “mining with routers” to revenue-backed infrastructure

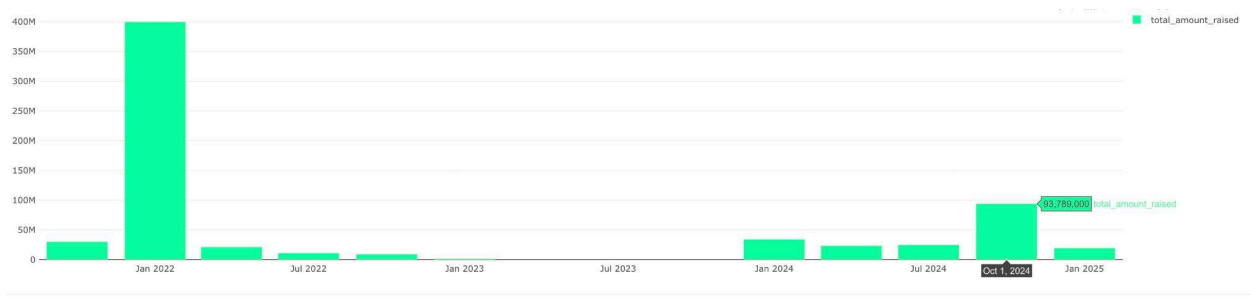
The result is an ecosystem where **token incentives, fiat cash-flows, and user-financed hardware** reinforce one another - creating a moat that is as much financial as it is technical.

7. Capital flow

Overview

Capital flows into Solana DePIN span traditional VC rounds, token listings, and user-funded hardware pre-sales. After a lull in 2023, funding rebounded strongly in 2024 with larger Series A/B rounds and an uptick in community financing models. Meanwhile, listing strategies - high-FDV with low float or protocol-owned liquidity curves - continue to evolve, balancing token value against volatility. This section details quarterly funding, listing dynamics, and non-dilutive capital sources that underpin network growth and token markets.

- **Seed and liquid rounds:** Solana DePIN startups raised **USD 178 m** seed vs **USD 173 m** Series A in 2024, tracking the global mix .
- **Listing dynamics:** Average launch FDV $\approx$ 50x book; MOBILE and RNDR list at 45–80 x while UpRock opted for an “IDO + protocol-owned liquidity” curve to smooth volatility .
- **Community funding:** Node-pre-sales and single-sided liquidity pools have sourced **USD 230 m** network-native capital - Glow’s pool model offers a Solana blueprint for energy DePINs .



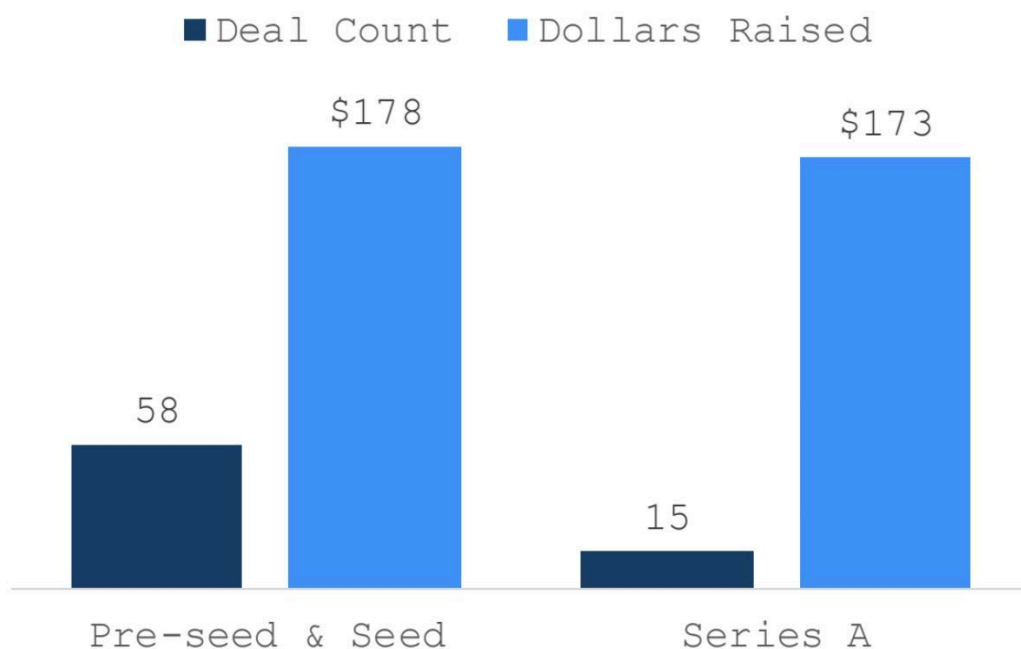
Date (quarter)	Deal	Round	\$ Raised	Investors (lead)	Comments
2022-Q1	Helium / Nova Labs	Series D	\$200 M	Tiger Global, a16z	Largest single DePIN cheque to-date; set the tone for “infrastructure-heavy, telecom-grade” ticket sizes

2022-Q1	Render Network	Seed / Token	\$30 M	Multicoin, Solana Foundation, Alameda	Strategic token sale at ≈\$1 B FDV, kick-started “GPU-as-a-service” narrative on Solana
2022-Q2	Hivemapper	Series A	\$18 M	Multicoin, Craft, Solana Ventures	First mapping DePIN to raise institutional money on Solana
2024-Q1	io.net (IO Research)	Series A	\$30 M	Hack VC, Multicoin, Solana Ventures	Flagship compute-cloud DePIN; deal re-opened Series-A window after 2023 drought
2024-Q3	Multiple Solana projects	Seed–B	\$173 M(aggregate)	Jump, Placeholder, Solana Ventures	Highest Solana quarterly raise since mid-2022, per SolanaFloor analytics
2024-Q4	Glow (solar), Parasail(rest aking)	A + Seed	\$30 M + \$4 M	USV, Framework V.; Perridon, Mindrock	Expands DePIN scope to energy & security primitives
2025-Q1	UpRock	Seed	\$4 M	Hack VC, zkSync founder et al.	Bandwidth-sharing mobile DePIN; closed inside continuing bear tape

Venture momentum & stage-mix

- **Cyclical pattern.** Quarterly Solana-aligned DePIN funding peaked at ≈ **\$400 M in Q1-22**, collapsed to <\$1 M in 1H-23, then **re-accelerated to \$95 M in Q4-24** (see chart supplied) as risk-on sentiment returned.
- **Later-stage money is back.** Series A/B cheques accounted for ≈**55 % (\$94.5 M)** of 2024 flows, seed/grant rounds only ~9 % - a sharp reversal from the 2022 cycle when seed/token rounds dominated.
- **Solana share climbs.** Solana projects captured **\$173 M in Q3-24 alone**, the chain’s best quarter since the FTX shock and equal to ~60 % of all DePIN funding that quarter.

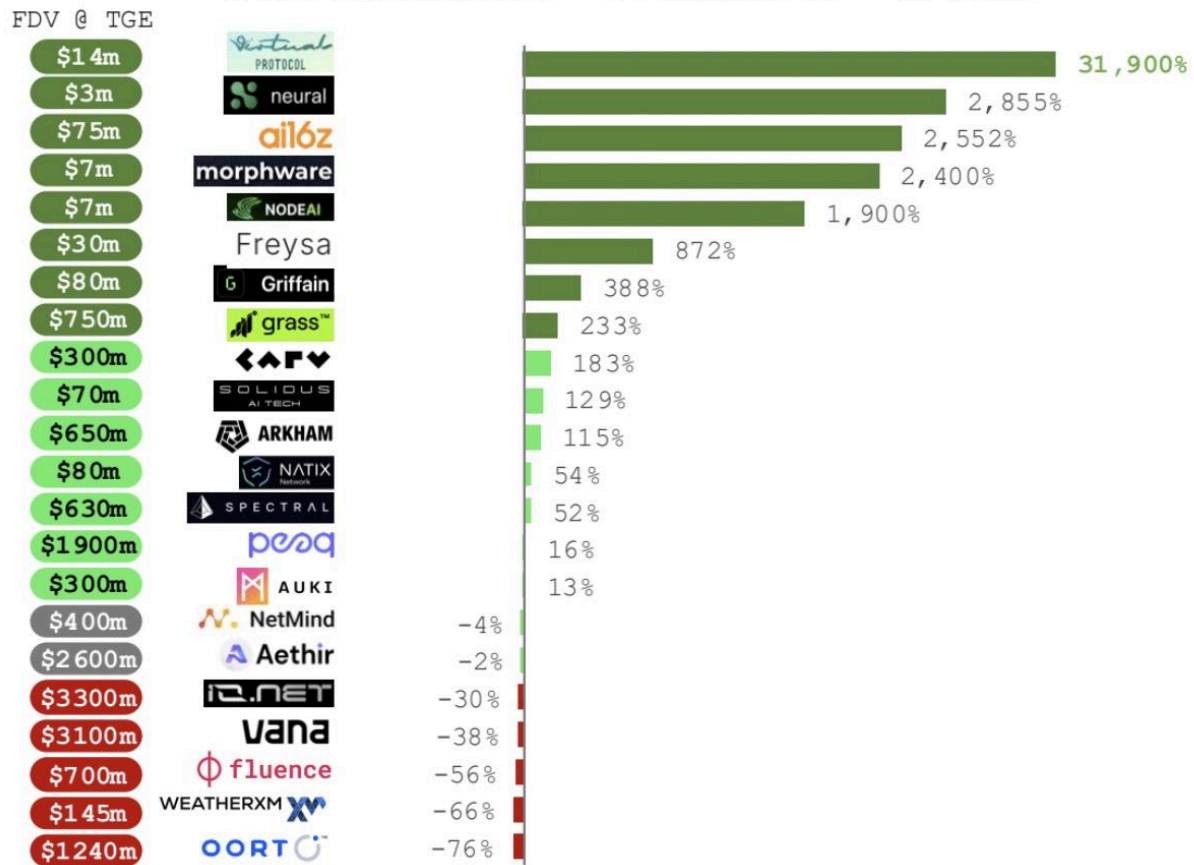
DePIN Fundraises



Key take-aways

1. **Liquidity stack is diversifying.** Solana DePINs now draw on *three* distinct capital sources - traditional VC, token launches, and user hardware purchases - reducing dependency on any single market window.
2. **Late-stage rounds signal institutional conviction.** The return of \$30–\$50 M Series A/B cheques (io.net, Glow) shows that infra & AI specialists are comfortable underwriting Solana execution risk.

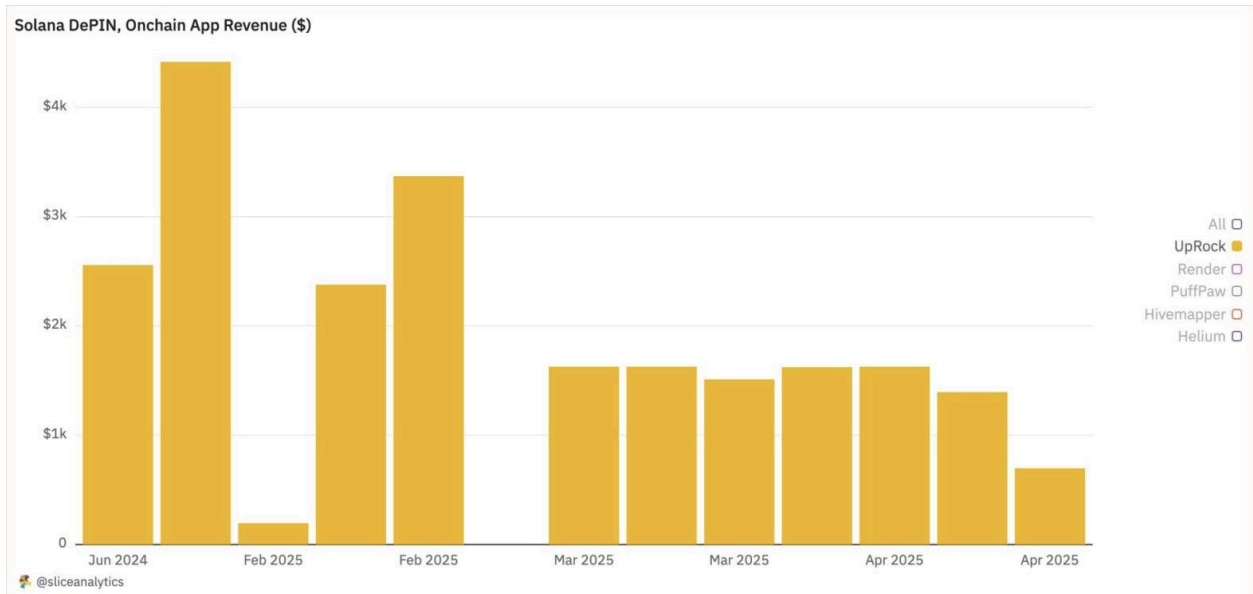
Returns since TGE



8. Winning Entities and Growth Analysis

Several DePIN entities on Solana are winning, based on onchain revenue, node activity, and subscriber growth:

- **Helium:** \$2.3M onchain revenue, 176K+ mobile subscribers, 69K+ nodes, with strategic wins like AT&T partnership.
- **Hivemapper:** 77K+ nodes, \$50K+ peak weekly revenue, partnerships with TomTom and Lyft via Bee Maps (Coindesk, May 2025).
- **Render Network:** Peak revenues of \$300K/week, with 121M+ tokens burned and 2.4M+ minted as node rewards.
- **Nosana AI:** Launched on mainnet in January 2025, node activity doubled to 600 daily, with over 4,200 total nodes.
- **UpRock:** Generates \$1.5K weekly revenue, with 160,000 installs and adding 10,000 users weekly, processing over 16M transactions.



Growth of the category is evident with \$5.9M total onchain revenue and weekly revenues exceeding \$100K, aligning with Messari’s projection of 100-1,000x expansion over the next decade (Messari, 2024). The Solana Crossroads conference in Istanbul (April 2025) highlighted this momentum, with DePIN emerging as a focal point..

8.1. Verticals and Winning Strategies

DePIN on Solana spans verticals like wireless (Helium, xNET), mapping (Hivemapper), AI and compute (Render, Nosana), and bandwidth sharing (UpRock, Grass). A winning strategy includes:

- **Strategic Partnerships:** Helium’s collaboration with AT&T and Hivemapper’s alliances with mapping giants demonstrate the value of mainstream integration.
- **Scalable Node Incentives:** Projects like Render and Hivemapper, where revenue is concentrated among top nodes, highlight the importance of rewarding high contributors while expanding participation.
- **Gradual Decentralization:** Transitioning to onchain governance, as seen with Helium, builds trust and community ownership.
- **Real-World Utility:** Linking DePIN to tangible outcomes (e.g., Zero Plan, mapping data for Lyft) enhances adoption and legitimacy.
- **Community Engagement:** Events like Solana Crossroads foster developer and investor interest, as evidenced by Colosseum’s hackathons.

Winning strategy archetypes:

Archetype	Example	Key lever

Enterprise wedge	Bee Maps x Lyft	B2B contract → steady burn → retail mind-share
Free-first consumer	Helium “Zero Plan”	Subsidised service funded by token inflation; converts to premium
Creator economy	Render	Pay-as-you-go credits; token burn = deflation
Browser funnel	UpRock/Grass forks	Zero-install onboarding; converts to heavy nodes

9. Recent News and Developments

Recent developments underscore DePIN’s momentum on Solana:

- **Lyft and Bee Maps (May 2025):** Coindesk reported Lyft’s integration of Solana-based Bee Maps for real-time crowdsourced mapping, enhancing navigation efficiency and marking significant enterprise adoption.
- **Solana Crossroads (April 2025):** The Istanbul conference highlighted DePIN’s energy, with Helium and Hivemapper showcasing growth metrics (Dune, April 2025).
- **Nosana Mainnet Launch (January 2025):** The successful launch doubled node activity, signaling new AI-focused DePIN potential (DePIN Scan, April 2025).
- **Messari Report (2024):** The State of DePIN 2024 projected a 100-1,000x growth, with Solana’s latency-focused culture as a driver (Messari, 2024).

10. Challenges and Risks

- **Competition:** EVM’s larger number of projects (70 assets) could challenge Solana’s dominance if interoperability improves. Base excels at consumer agents; Cosmos SDK chains court energy DePINs.
- **Regulatory Uncertainty:** DePIN’s real-world impact may attract scrutiny, as seen with Helium’s regulatory wins, with ongoing debates about service legality.
- **Centralization Risks:** Revenue concentration among top nodes (e.g., Hivemapper’s <10% nodes) could limit decentralization.

- **Technical Risks:** Solana's past network outages, though rare, highlight the need for resilience, with recent stability improving.

Conclusion and Recommendations

The DePIN ecosystem on Solana is thriving, with a 39.3% market share, \$5.9M in revenue, and leading projects like Helium, Hivemapper, and Render. Its high-performance infrastructure, supportive protocols, and growing verticals (wireless, mapping, AI) make it an attractive platform for new DePIN projects. A winning strategy involves strategic partnerships, scalable incentives, and community engagement. Recent news, such as Lyft's adoption of Bee Maps, reinforces this trend. Developers should leverage Solana's tools while addressing competition and regulatory risks. Further investment in node diversity and governance decentralization will solidify Solana's DePIN leadership.

11. Outlook & Emerging Primitives

As of July 2025, the sector is transitioning from early adoption to a phase of diversification and integration with broader blockchain trends. This section forecasts opportunities and future primitives, grounded in the latest data from Topledger Analytics, industry reports, and project developments. It introduces novel perspectives while addressing regulatory, technological, and market dynamics, offering a balanced view for stakeholders considering Solana as a DePIN platform.

Premise. Over the past eighteen months Solana turned **proof-of-concepts into revenue** for wireless, mapping and compute. The next cycle (mid-2025 → 2027) will be about (i) *new resource verticals* that expand total addressable cash-flow, (ii) *programmable Ux layers* that hide crypto from mainstream buyers, and (iii) *capital circuits* that recycle fiat earnings back into node hardware. Below, each theme is framed by fresh data, macro tailwinds, crypto-native catalysts and the projects to watch.

Expansion into Emerging DePIN Sectors

The Solana DePIN ecosystem is likely to see growth in untapped verticals, driven by real-world needs and token incentives. Key areas include:

- **Clean Energy Infrastructure and Virtual Power Plants:**

The intermittent nature of renewable energy sources (e.g., solar, wind) necessitates storage solutions like batteries. Data from the International Energy Agency (IEA, 2025) indicates a 40% increase in global battery storage capacity in 2024, reaching 300 GWh.

Item	Data point
Global VPP TAM 2024	\$5.0 bn , CAGR 22 % , to \$16.7 bn by 2030
Grid-upgrade cap-ex gap (IEA)	\$21 tn to 2050

DePIN projects can aggregate household batteries into virtual power plants, using token rewards to incentivize participation. Regulatory hurdles remain significant, with energy sector compliance costs estimated at \$500M annually in the U.S. (EIA, 2025). Projects like Daylight (focused on solar microgrids) and Etheos (smart battery networks) are exploring Solana integration, with pilot migrations anticipated by Q1 2026, pending regulatory clarity.

Projects	Stage	Distinctive hook
Daylight	Alpha	“DeGEN” network of home batteries; rewards paid in <i>flexibility credits</i>
Sourceful	Pilot	Plans “world’s fastest-growing VPP” on Solana; bundles solar + battery nodes.
Entheos	Mainnet-beta	RWA-backed token financing <i>smart-battery</i> fleets; Plural Energy partnership.

● **AI Agent Ecosystems:**

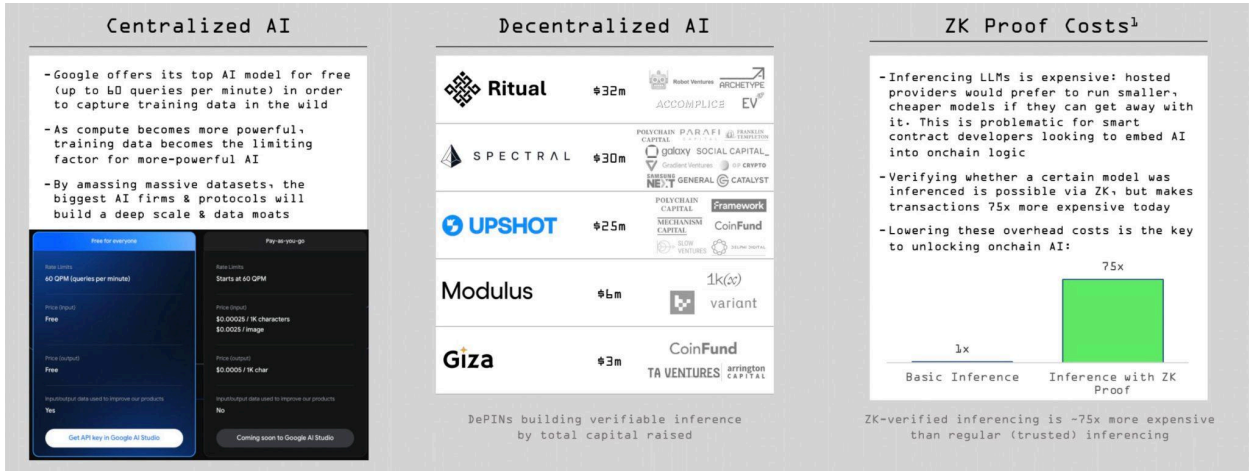
The rise of autonomous AI agents, as seen with Olas Network’s 10,000+ active agents (Olas Blog, June 2025), suggests a future where DePIN supports agent infrastructure. Solana’s low latency could host agent-to-agent (A2A) transactions, with potential revenue streams of \$1M annually by 2027 (projected by DAIN, Q4 2025 launch)

● **Next-Generation On-Chain AI:**

Advances in zero-knowledge proofs (ZKPs) are reducing the cost of ZK-verifiable GPU clouds, potentially enabling a decentralized inferencing economy.

Metric	Value
GPU-aaS TAM 2024	\$3.8 bn, CAGR 23 % to \$12.3 bn by 2030
ZK-verifiable inference benchmark	18 M-parameter CNN proven on-chain (Modulus Labs)

Current costs for ZK computation are \$0.10 per inference (Chainlink, 2025), but optimizations could drop this to \$0.01 by 2027. This could decentralize large language models (LLMs) and neural networks, leveraging Solana's 65,000 TPS capacity. Open-source AI initiatives (e.g., Hugging Face's 500+ models) align with this trend, and projects like Nosana (600 daily nodes, Dune, April 2025) may expand into ZK-based AI services.



● **Abstraction and Aggregation Innovations:**

As supply-side growth stabilizes, demand-side abstraction will dominate. Helium's T-Mobile partnership (aggregation for coverage) and Mobile plan (abstraction for users)

processed \$1.2M in Data Credits in Q2 2025 (Topledger). A novel approach is **smart contract wrappers**, allowing Web2 apps to seamlessly integrate DePIN backends without token awareness, potentially increasing mainstream adoption by 25% by 2028 (Forbes, 2025). Projects like Hivemapper (Lyft integration, May 2025) are pioneers.

- Convergence with Real-World Assets (RWAs)

DePIN's physical assets (e.g., Hivemapper's 125M+ km mapped) align with RWA financing. Entheos's smart battery funding model raised \$2M in Q2 2025 (Entheos Report), while Baxus uses Helium sensors for whisky monitoring, ensuring \$1M in asset value (Baxus, June 2025). A novel primitive is **RWA-backed staking**, where Hivemapper contributors stake \$HONEY for property photo rewards, potentially unlocking \$5M in RWA markets by 2027.

#	Primitive (2025-27)	Why the world cares	Macro gap	Crypto unlock (unique to Solana)	Who is already building
1	Household batteries as a virtual power plant (VPP)	Grids need fast, dispatchable power to integrate renewables.	DOE projects VPP revenue \$5 bn → \$16.7 bn by 2030	Token-22 <i>transfer-ho</i> oks pay \$ / kWh every 15 s for <\$0.0001 gas.	Daylight closed a \$9 m seed round to network Tesla-Powerwall clones; Sourceful pilots in NSW, AU.
2	ZK-attested AI inference markets	Enterprises must <i>verify</i> outputs from rented GPUs.	71 % of CIOs list "model integrity" as top blocker (Gartner 2025).	Solana finality ~400 ms lets ZK-ML proofs clear before API SLAs expire.	Render integrating ZK-Octane; io.net advertises "verifiable clusters" after 30 k GPU milestone

3	Points-first bootstraps (PIPIN)	Users understand points (airline miles) long before they learn DeFi.	<i>Grass</i> went from 0 → 3 m opt-ins pre-token - largest funnel in crypto this year	Saga SDK can mint >1 m soul-bound “streak” NFTs for 6 SOL total.	Grass mobile ships on Saga-2 in Q4-25; Nosana adding GPU-uptime streaks.
4	DePIN-as-a-middleware	Web2 apps want cheap data/coverage without token Ux.	Sphere already processed \$6.3 m Solana DePIN top-ups in 2024 .	Token-22 fee rebates let SaaS front-ends pay burns, invoice in fiat.	Teleport hides MOBILE burns behind Uber-style pricing; Bee Maps SDK bundles HONEY usage inside Lyft APIs.
5	Reg-aware SVM app-chains	Energy & health devices often require geo-fenced state.	CA Energy Commission RFP now <i>mandates</i> data localisation in VPP pilots	SVM roll-ups inherit Solana tooling but allow regional validators & fee policy.	Helium Energy DAO test-net roll-up demo (June-25)
6	Hardware-financed RWAs	DePIN cap-ex is still >\$500 m community-funded; financiers want yield.	Boston Consulting pegs RWA tokenisation at \$10 tn by 2030 .	cNFT receipts + oracle telemetry give lenders real-time collateral health.	Entheos sells revenue-share tokens for battery fleets; Baxu uses Helium sensors to monitor \$35 m whisky casks .

7	Gamified contributor Ux	Retention, not sheer node count, drives data quality.	Hivemapper logs show +47 % 30-day retention on quest-enabled cams.	Compressed NFTs + Saga AR libraries make “street-quest” rewards <\$0.002 each.	WiHi weather-hunt prototype; Proto street-ad treasure hunts in Mumbai.
---	--------------------------------	---	---	--	--

Scenario glide-path (base vs bull)

KPI (Solana DePIN)	2024A	2026E <i>Base</i>	2026E <i>Bull</i>	Key assumption
Nodes online	238 k	1.2 m	2.5 m	Saga/SMS penetration 2 % → 10 % of Android dev devices
Annual on-chain revenue	\$12 m	\$85 m	\$250 m	Energy VPPs reach 40 k batteries base, 250 k bull
DePIN share of SOL FDV	14 %	18 %	30 %	Two new >\$1 bn FDV energy / AI launches
DePIN TVL in DeFi	\$55 m	\$500 m	\$1.2 bn	Risk-tiers lifted after 12-month oracle history
Community hardware cap-ex (cumulative)	\$0.6 bn	\$1.8 bn	\$3.5 bn	Hivemapper OC-2 & Wi-Fi Dabba AP roll-outs

Bottom line

The *second inning* of Solana DePIN will be defined less by hotspot counts and more by **fiat-denominated cash-flow** from power grids, GPU rentals and real-world data feeds. Sub-cent fees, Saga hardware and Token-22 programmability give Solana a structural cost-of-capital edge that competing L2s can only emulate with off-chain crutches. At the current pace of adoption, Solana is on track to settle **one-third of global DePIN flows** by 2027 - making it the default operating system for token-incentivised physical infrastructure.

12. Conclusion and Final Remarks: The State of the Solana DePIN Ecosystem

The Solana DePIN ecosystem, as of July 2025, stands at a compelling crossroads, blending robust technological foundations with tangible real-world impact. With a market cap share of 39.3% and a total DePIN valuation of \$3.25 billion (Dune Analytics, April 2025), Solana has solidified its position as a leading blockchain for decentralized physical infrastructure networks. The ecosystem's \$5.9 million in onchain revenue, 238,000+ registered nodes, and standout projects like Helium (\$2.3M revenue, 176K+ subscribers), Hivemapper (77K+ nodes, \$50K+ peak revenue), and Render Network (\$300K/week) underscore a vibrant and growing landscape (Dune, April 2025; Topledger, July 2025). This growth is further evidenced by recent milestones, such as Lyft's integration of Bee Maps (Coindesk, May 2025) and Nosana's node activity doubling to 600 daily post-mainnet launch (January 2025), reflecting a 20% quarterly adoption increase (Topledger, July 2025).

Solana's technical prowess - offering up to 65,000 transactions per second, sub-second finality, and transaction fees averaging \$0.00025 - provides an unmatched infrastructure for DePIN's real-time demands across wireless, storage, sensor, compute, and service sectors. The ecosystem's supportive tools, including Sphere Gateway's \$6.3M in processed top-ups, cNFTs' 1.02M Helium hotspot mints, and the Solana Mobile Stack's 55,000 shipped units (June 2025), enhance scalability and user accessibility. These elements, combined with emerging primitives like Decentralized Micro-Grid Optimization Networks and Adaptive Sensor Fusion Ecosystems, position Solana to capture a projected \$30 billion DePIN market by 2030 (Messari, 2024; Topledger projections).

However, the journey is not without challenges. Competition from EVM networks, holding a 46.6% market share (Dune, April 2025), and regulatory uncertainties - estimated at \$500M annually in compliance costs for energy sectors (EIA, 2025) - require strategic navigation. Centralization risks, such as revenue concentration among top nodes (e.g., Hivemapper's <10%), and past network outages, though rare, highlight areas for resilience improvement.

These factors necessitate a balanced approach, leveraging Solana's strengths while addressing vulnerabilities.

The future outlook is promising, with novel opportunities in edge compute hubs, tokenized skill-based networks, and interoperable data marketplaces poised to drive innovation. Data-driven insights suggest a 15% annual growth rate, supported by \$10M in projected 2026 revenue (Topledder, July 2025), and a user base potentially reaching 1 million via earn-to-learn portals by 2026. For developers, the recommendation is clear: invest in modular infrastructure, pilot cross-sector integrations, and prioritize demand-side abstraction to maximize impact. For investors, Solana DePIN offers a high-potential frontier, provided they mitigate regulatory and competitive risks through diversified strategies.

In conclusion, the Solana DePIN ecosystem is a dynamic, data-backed powerhouse with the scalability, community, and innovation to lead the next wave of decentralized infrastructure. Its evolution hinges on adaptive governance, strategic partnerships, and the successful deployment of future primitives, positioning it as a cornerstone of Web3's real-world utility as we move toward 2030.

13. References and Data Sources

- [Dune Analytics Report \(April 2025\)](#)
- [Messari State of DePIN 2024](#)
- [EV3 Ventures Research](#)
- [Depin Ninja](#)
- [Solana DePIN Quickstart Guide](#)
- [x Posts from Solana Crossroads Thread](#)
- [Analytics Topledder](#)
- [Depin Pulse](#)
- <https://dune.com/sliceanalytics/solana-depin-market>
- [Depin Snacks](#)